University of Massachusetts Dartmouth

Department of Mechanical Engineering

Computational fluid dynamics in OpenFOAM for passive control of flow separation on NACA 63-018 airfoil using vortex generators and leading edge tubercle

A Thesis in

Mechanical Engineering

by

Milad Rakhsha

Copyright 2012 by Milad Rakhsha

Submitted in Partial Fulfillment of the
Requirements for the Degree of
Master of Science

August 2015

Abstract

Computational fluid dynamics in OpenFOAM for passive control of flow separation on NACA 63-018 airfoil using vortex generators and leading edge tubercle

by Milad Rakhsha

Vortex generators and leading edge tubercles are known as passive geometries, which can be used to control flow separation over airfoils. In the present research, computational fluid dynamics (CFD) analysis is conducted in order to study the effect of vortex generator (VGs) and leading edge tubercles (LETs) on NACA63018. As far as VGs are concerned, double-sided VGs are investigated for the sole purpose of very high solidity hydro-kinetic vertical axis turbines, whose blades operate predominantly in the stall region at low tip speed ratios.

To this aim, OpenFOAM is used as the CFD package to model the flow over the airfoil in pre-stall and stall regions. The structured parametric mesh created in blockMesh dictionary provides a powerful framework to study the effect of VGs geometry and their location parametrically. Similarly, parametric analysis is applied to the wavelength and amplitude of leading edge tubercles. For the Re numbers of interest in this study, $Re = 3 \times 10^6$ and $Re = 5 \times 10^5$, the $k - \omega SST$ turbulent model is used, which offers the best accuracy in comparison to other turbulent models. The problem is modeled using the steady-state solver, simpleFoam, over a range of angle of attacks from 0° to 18°, throughout which the flow behaves as pseudo-steady. The effect of VGs and LETs geometries are configured optimally at one high angle of attack. The optimal configurations are fixed for the rest of the angle of attacks at both high and low Re numbers.

The optimal position of non-submerged VGs are found to be at 10% and 20% of the chord length from the leading edge for the low and the high Re numbers respectively. Results show that applying double sided VGs shows an increase in lift-to-drag ratio

in the stall and post-stall regions. However, applying LETs does not improve the performance of the airfoil for the cases studied in the present study.

Keyword: Passive flow control, Vortex generator, Leading edge tubercles, Vertical Axis Turbine, NACA63018, OpenFOAM

Acknowledgements

Foremost, I would like to express my sincere gratitude to my supervisor and my teacher, Professor Raymond Laoulache, for his continuous support, guidance, intellectual wisdom and patience during this study.

I place on record, my sincere thank you to the Department's faculty members Professor Amit Tandon and Professor John Rice for what they taught me in Fluid and Continuum Mechanics and for their support.

I would also like to express my gratitude to Ms. Susan Cunha, department secretary, for her kind help and Mr. Andrew Smart, the administrator of the Linux server, for his technical support.

I would like to thank my colleague, Behrooz Amirzadeh, who shared with me his valuable knowledge in bash scripting, LaTeX and Visit post-processing.

I thank my family members for the unceasing encouragement, support and attention throughout this venture.

Contents

List of Figures **viii**

List of Tables **xi**

Nomenclature **xii**

1 Introduction **1**

1.1 Flow Control 1

1.2 Vertical Axis Turbines 3

2 Basic Concepts **7**

2.1 Drag components 7

2.2 Stall and Flow control 7

2.3 NACA airfoils 9

2.3.1 Four-digit series 10

2.3.2 6-series airfoils and NACA $63_3 - 018$ 10

3 Literature Review **12**

3.1 Vortex Generators 12

3.2 Leading Edge Tubercle 14

3.3 Computation Fluid Dynamics 16

4 Computational method **18**

4.1 Meshing Technique 19

4.1.1 2D Mesh 19

4.1.2 3D Mesh with VG 21
4.1.3 3D Mesh with Undulating Leading Edge 22
4.1.4 Mesh Quality 24
4.2 Governing equations 27
4.3 Boundary Conditions 29
4.4 Finite Volume Schemes 33
4.5 Solution Control 34

5 Results **36**
5.1 Validation of numerical results 36
5.2 Vortex generator 39
5.3 Undulating leading edge 48

6 Conclusion and scope of future work **55**

Appendices **55**

A SIMPLE Algorithm in OpenFOAM **56**
A.1 Basic information 56
A.2 Mathematics of SIMPLE algorithm 56
A.3 OpenFOAM source code 61

B An OpenFOAM Tutorial **65**
B.1 Structure of the case 65
B.2 Files 65

Bibliography **65**

List of Figures

1.1 Derivation of angle of attack ,α, as a function of angle of rotation, θ. 4

1.2 Variation of angle of attack vs. angle of rotation of a blade at different tip speed ratios for $\overline{f} = 0.3$. 6

2.1 Occurrence of stall in airfoils 8

2.2 Occurrence of stall in higher angle of attacks in modified airfoils . . . 9

2.3 Schematic diagram of NACA airfoils 10

3.1 Counter rotating(left) and Co-rotating (right) VVGs 13

3.2 Undulating leading edge of an airfoil inspired by humpback whale . . 14

4.1 Fluctuation in the lift and the drag coefficients VS. iterations for $\alpha = 16^\circ$ and $Re = 3 \times 10^6$ using simpleFoam. 19

4.2 Time history of the lift and the drag coefficients at $\alpha = 16^\circ$ and $Re = 3 \times 10^6$ using pimpleFoam 20

4.3 Schematic of the outer O-grid and the inner C-grid blocks and grid spacing parameters 21

4.4 Structured mesh of 2D airfoil 22

4.5 Zoomed view of structured mesh around the NACA6301 airfoil 23

4.6 Zoomed view of the surface mesh around the VG 23

4.7 Surface Mesh of 3D airfoil with leading edge tubercle 24

4.8 Hexahedral, tetrahedral, prism and pyramid schematics 25

4.9 Skewness definition in OpenFOAM 26

4.10 Top view schematics of the airfoil for 3D simulations with VGs. The half model (Left) and the full model (Right) 31
4.11 Top view schematic of the 3D airfoil with undulating leading edge . . 32
4.12 Absolute residuals of variables at $\alpha = 8^\circ$ and $Re = 5 \times 10^5$ 35

5.1 Comparison of the experimental results and CFD prediction of the present study at $Re = 3 \times 10^6$. 38
5.2 Comparison of the velocity profiles at $\alpha = 16^\circ$ between the bare airfoil and the one with VG 41
5.3 Velocity profiles (m/s) at different sections downstream the flow . . . 42
5.4 Vorticity profiles at different sections along the spanwise direction . . 43
5.5 Time evolution of the vorticity profiles at $\alpha = 16^\circ$ and $Re = 3 \times 10^3$ for the bare airfoil with characteristic time scale $t_s = c/U_\infty = 0.33$ . . 44
5.6 Time evolution of the vorticity profiles at $\alpha = 16^\circ$ and $Re = 3 \times 10^3$ for the 3D airfoil with VGs. Sections at $\Delta z/d' = 4/6$ 45
5.7 Effect of using single-sided vortex generators on top and double-sided vortex generators on top and bottom sides at $Re = 3 \times 10^6$ 46
5.8 Effect of using double-sided vortex generators on the suction side and the pressure side at $Re = 5 \times 10^5$ 47
5.9 Comparison of lift to drag ratio between the bare airfoil and the passive airfoil with double-sided VGs on both suction and pressure sides . . . 48
5.10 Top and front view schematics of the 3D airfoil with undulating leading edge 49

5.11 Comparison of the velocity profiles (in m/s) at $\alpha = 18°$ between the bare airfoil and the modified airfoil with leading edge tubercles with $\overline{\lambda} = 0.1$ and $\overline{A} = 0.0125$ at a section 10% of the chord from the leading edge 50

5.12 Vorticity profile at $\alpha = 18°$ for the modified airfoil with leading edge tubercles with $\overline{\lambda} = 0.1$ and $\overline{A} = 0.0125$, at a section 10% of the chord from the leading edge 51

5.13 Comparison of the velocity profiles at $\alpha = 18°$ between the bare airfoil and the modified airfoil with leading edge tubercles 52

5.14 Effect of using leading edge tubercles with $\overline{\lambda} = 0.1$ and $\overline{A} = 0.0125$ at $Re = 5 \times 10^5$. 54

B.1 Structure of the OpenFOAM case 66

List of Tables

4.1 Boundary conditions of the CFD model 1)at high Re, 2)at low Re, 3)with VG and 4)with leading edge tubercle 32

4.2 Mathematical definition of BCs used in OpenFOAM 33

5.1 Grid independence study parameters 36

5.2 Effect of the vortex generator geometrical parameters on the lift and the drag coefficients at AOA=16° 39

5.3 Effect of the tubercles' control parameters on the lift and the drag coefficients at AOA=16° and $Re = 5 \times 10^5$ 53

Nomenclature

b	bare airfoil
B	number of VAT blades
c	reference airfoil chord
C_d	drag coefficient
C_l	lift coefficient
d	distance between vortex generators
D	spanwise spacing of vortex generators pairs
D_{VAT}	diameter of the VAT blades
h	height of vortex generator
l	length of vortex generator
Re	Reynolds number ($= \frac{\rho U_\infty c}{\mu}$)
VG	vortex generator
w	width of the airfoil into the paper
Δx	position of VG from the leading edge
y^+	y value in law-of-the-wall variable $u_* y/\nu$
u_*	friction velocity, $\sqrt{\tau_w/\rho}$
GAMG	generalised geometric-algebraic multi-grid
PCG	preconditioned conjugate gradient

PBiCG	preconditioned bi-conjugate gradient
DIC	Diagonal Incomplete-Cholesky
DILU	Diagonal incomplete LU

Greek symbols

α	angle of attack in degrees
β	yaw angle between a pair of vortex generators
δ	boundary layer thickness
μ	dynamic viscosity of fluid
ρ	density of fluid
γ	yaw (skew) angle between VG obstacle
λ	tip speed ratio ($= \Omega R/U_\infty$)
σ	solidity ($= Bc/\pi D_{VAT}$)

Subscripts

t	turbulent
w	wall condition
∞	free-stream value

Superscripts

$\bar{}$	average value
0	initial
$*$	interim value

Chapter 1

Introduction

Engineers have always been looking for renewable sources of energy. Among all the these sources, wind energy has always been one of the most attractive ones. Therefore, wind turbines have been developed and modified in a great number of ways to exploit wind energy as much as possible. Among the different forms of wind turbines, Vertical Axis Turbines (VAT) were developed because of the advantageous they offer over Horizontal Axis Turbines (HAT), including but not limited to being insensitive to the wind direction. However, just like any other design, VATs performance are deteriorated in certain conditions. One of the most important factor that prevents the VATs from fully exploiting the wind energy is flow separation from the blades installed on the turbine. Therefore, flow field control becomes essential.

1.1 Flow Control

Flow field control around different objects which results in reduction of drag coefficient and/or augmentation of lift coefficient is of great technological importance. By reducing the drag coefficient billions of dollars in fuel costs for land, air and sea vehicles, will be saved, not to ignore the impact on the performance of renewable energy devices [11, 20, 12]. The important point here is that these objectives are not necessarily mutually exclusive. Therefore, the challenge in selecting the flow control method is to find the desirable goals at minimum cost and without adversely affecting

the other goals. Generally speaking, eliminating or at least delaying flow separation which results in a reduction in pressure friction, is the main objectives of flow control. This, however, as it will be shown later, does not guarantee that other criteria will not be deteriorated. For instance, by applying vortex generators on airfoils friction drag increases. On condition that separation does not occur, the form drag does not have a dominant effect on the total drag. Therefore, applying VGs will adversely affect the drag. This associated drag penalty is noticeable before the stall region, hence. Consequently, depending on the application that is of interest, a specific flow control method may or may not be worth using.

Different classifications are known for flow control. Whether the technique is applied at the wall or away from it, could be named as one these classification. One can control the flow by use of surface roughness, shape or curvature or even by rigidly moving the surface. The flow could also be controlled by use of devices placed away from the wall. Large-eddy breakup devices (outer-layer devices, or OLDs), acoustic waves bombarding a shear layer from outside are some of the examples of this technique. In addition, based on the expenditure of energy, another classification is defined as active or passive flow control. In active control an external source of energy is required to maintain the desirable effect while in order to passively control the flow no input energy is required [12]. The later approach is specially suitable for underwater small VATs where active control of the system becomes even more costly [5]. Active method itself is divided into reactive or predetermined. Reactive methods use a continuously adjusting input using either open, feedforward or feedback loop while predetermined control includes the control devices to input energy regardless of the particular state of the system. Another important classification of flow control is to consider if the control devices directly change the shape of velocity profile or selectively influence the dissipative eddies [12].

In the present study, two passive techniques are used to control the flow separation over NACA 63-018; vortex generators and leading edge tubercles.

1.2 Vertical Axis Turbines

In order to use vortex generators in VATs, it is essential that two pairs of vortex generators are placed on the pressure side and the suction side of the blade as the turbine develops lift not only in the upwind region, but also in the downwind region. In short, as blades of a VAT go around the vertical axis, they develop lift from both the inner side and the outer side. Thus experiencing the separation not only from the inner side but also from the outer side. This is schematically shown in figure 1.1. It can be seen that the inner side of the blade shown in this figure is the suction side in half of the cycle and becomes the pressure side in the other half.

Assuming that θ is the angle of rotation of a blade around the vertical axis (Azimuthal angle) as is shown in figure 1.1, the relation between α and θ is derived as follows.

The velocity seen by the airfoil can be written as:

$$\boldsymbol{U_r} = \boldsymbol{U} - \boldsymbol{U}_b \tag{1.1}$$

Where $|\boldsymbol{U}_b| = R\Omega$ is the absolute velocity of blade and U is a fraction of incoming velocity U_∞. Based on figure 1.1, $\mathbf{U_r}$ is written as:

$$\mathbf{U}_r = -\Big(R\Omega + Ucos\theta\Big)\hat{e}_\theta - \Big(Usin\theta\Big)\hat{e}_r \tag{1.2}$$

Angle of attack is the angle between $\mathbf{U_r}$ and $\hat{e}_\theta$:

$$\alpha = tan^{-1}\Big(\frac{Usin\theta}{R\Omega + Ucos\theta}\Big) \tag{1.3}$$

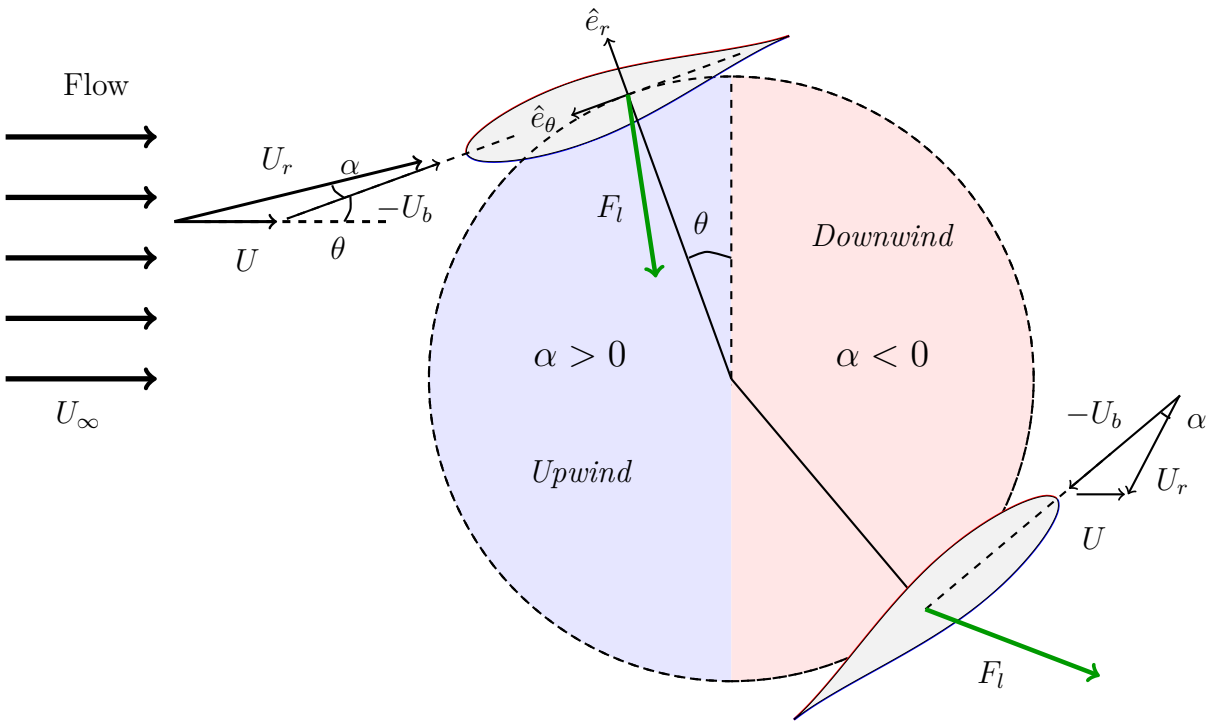


Figure 1.1: Derivation of angle of attack ,α, as a function of angle of rotation, θ. Blue side represents suction side which experiences flow separation.

Which can be written as:

$$\alpha = tan^{-1}\left(\frac{sin\theta}{R\Omega/U + cos\theta}\right) \tag{1.4}$$

Where $TSR = R\Omega/U_\infty$ and based on the defenition of induction factor, $f = 1 - \frac{U}{U_\infty}$, which is the fractional decrease in wind velocity between U_∞ and U, U in equation 1.5 can be written as $U = U_\infty(1 - f)$ and ultimately equation 1.5 is written as follows:

$$\alpha = tan^{-1}\left(\frac{sin\theta}{TSR/(1 - f) + cos\theta}\right) \tag{1.5}$$

It can be seen that for $0° < \theta < 180°$, α is positive and when $180° < \theta < 360°$, α becomes negative. Therefore the blades experience suction and pressure on each side

and double sided VGs are required to improve the performance of the airfoil in the whole cycle.

In the above equations f is a function of azimuthal angle θ, solidity and tip speed ratio. To demonstration the concept though, $\overline{f} = 0.3$, which is the typical average value of f for high solidity and low tip speed ratios [27, 39], is used to plot the variation of θ vs. α for different TSRs in figure 1.2. Figure 1.2, and also figures from other references such as reference [8], show that blades work in higher angles as tip speed ratio is decreased. It also can be seen, the lower the tip speed ratio, the higher the fraction of the time/angle-of-rotation that blades work in higher angle of attacks. Moreover, for low tip speed ratios, the secondary effects on the performance of the turbine are negligible and dynamic stall to which the lift and the drag coefficients of the blade are very sensitive, is the most important factor on the performance of the turbine. Besides, greater dynamics stall region is observed for laminar cases [27].

In addition, assuming that the airfoil works in the stall region at $\alpha > 14$, it can be concluded from figure 1.2 that the blades work more than 70% of the time in stall and post-stall regions for $TSR = 1.2$. This is the typical behavior of the VAT with $1 < \lambda < 2$.

Shiono et al [34] developed a Darrieus type vertical axis turbine suitable for tidal current power generation. They investigated the effect of solidity and number of blades on the performance of the turbine. Their results indicate that for a high solidity turbine with $\sigma = 0.366$, the maximum efficiency of the turbine is obtained at tip speed ratios around $TSR = 1.25$.

This work is aimed to improve the performance of the heavily loaded airfoils encountered in such hydro-kinetic vertical axis turbines with low tip speed ratio, TSR, and high solidity,σ. As it was stated, blades of such turbines, work predominantly in high angle of attacks and hence the stall and post stall regions . Thus, this work is targeted at

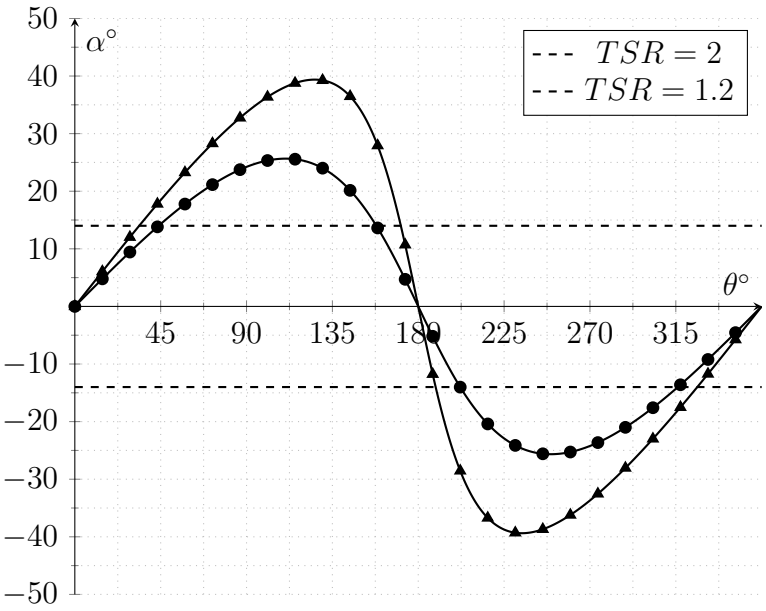


Figure 1.2: Variation of angle of attack vs. angle of rotation of a blade for $\overline{f} = 0.3$ at different tip speed ratios.

improving the performance of such blades at high angle of attacks.

This dissertation is outlined as follows. In chapter 2, basic concepts are investigated regarding the aerodynamics of wind turbine blades, and specifications of NACA airfoils and specifically NACA 63-018 foil which is of interest in the current study. In chapter 3 a review of the literature of VGs, LETs and computational fluid dynamics of VGs are presented. In chapter 4, the numerical method used to obtain the results are presented. In this chapter, the preprocessing part of the CFD problem is explained, governing Navier-Stokes equations and the RANS modeling to resolve the turbulence is explained. Finite volume discretization schemes and solution control are discussed as well. In cahpter 5, results of the CFD simulations are presented; validation of numerical results is presented and results of flow control using each of the methods discussed earlier is described. Finally, in chapter 6, it is concluded whether or not each method can be used and the scope of future studies are presented.

Chapter 2

Basic Concepts

2.1 Drag components

The drag force which airfoils experience while moving in fluids is usually divided into two components: Frictional drag and pressure drag. Frictional drag comes from the effect of boundary layer and existence of viscose flow. On the other hand, pressure drag comes from motion of eddies that are created in the fluid by the passage of bodies. This drag is associated with the formation of wakes behind the passing objects in fluids. Pressure drag is rather less sensitive to Reynolds number of flow than frictional drag. Frictional drag is important in attached flow when there is no separation. However, pressure drag is important for separated flow. Pressure drag is also known as form drag while friction drag is called viscose drag. If the drag force is dominated by viscose drag the body is called streamlined and when the drag is dominated by pressure drag the body is called bluff [1].

2.2 Stall and Flow control

In aerodynamics stall is defined as a condition when the lift coefficient does not increase by increasing the angle of attack and in contrast it begins to decrease. Another definition describe stall as a state when the lift at a particular angle of attack is lower than that of a lower angle of attack. This is schematically shown in figure 2.1.

The angle at which the behavior of lift coefficient changes is known as critical angle

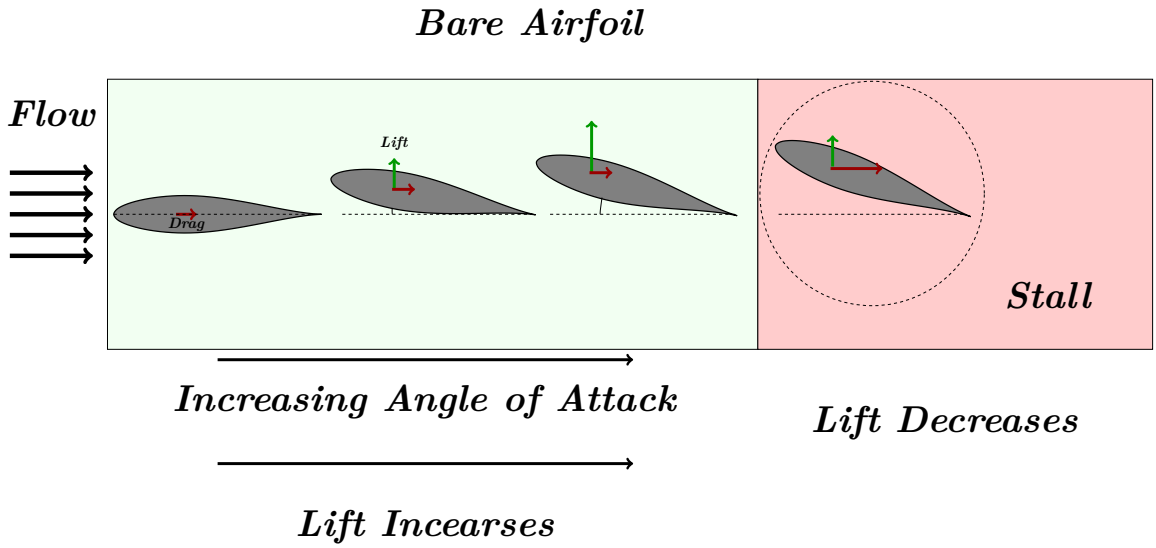


Figure 2.1: Occurrence of stall in airfoils

of attack. This angle depends on different geometrical and physical parameters such as the geometry of the airfoil, the roughness of the surface, Reynolds number and so on. The concept of stall is coupled with the concept of boundary layer separation, and boundary layer separation happens because of flow reversal due to the adverse pressure gradient which is imposed by the outer potential flow. The separation of boundary layer results in augmentation of displacement thickness and therefore augmentation of the drag coefficient[13, 4, 6]. At small angle of attacks, both of the top and the bottom surfaces of an airfoil experience relatively low pressure gradient and the flow is usually (depending on the airfoil geometry) attached to the surface over the entire length of the airfoil, yet, as the angle of attack is augmented the pressure gradient over the airfoil becomes larger and the adverse pressure gradient on the top rear portion of the airfoil becomes stronger, hence the flow separates and the wake region becomes greater, the pressure drag due to the formation of eddies increases and thereby the total drag increases. The airfoil is called stalled in this situation [1].

In order to alleviate the occurrence of stall that has been shown in figure 2.1, the

flow can be controlled so that it does not get separated from the airfoil. Figure 2.2 schematically shows how flow control improve the performance of the airfoils; the angle of attack can be increased without going into stall region.

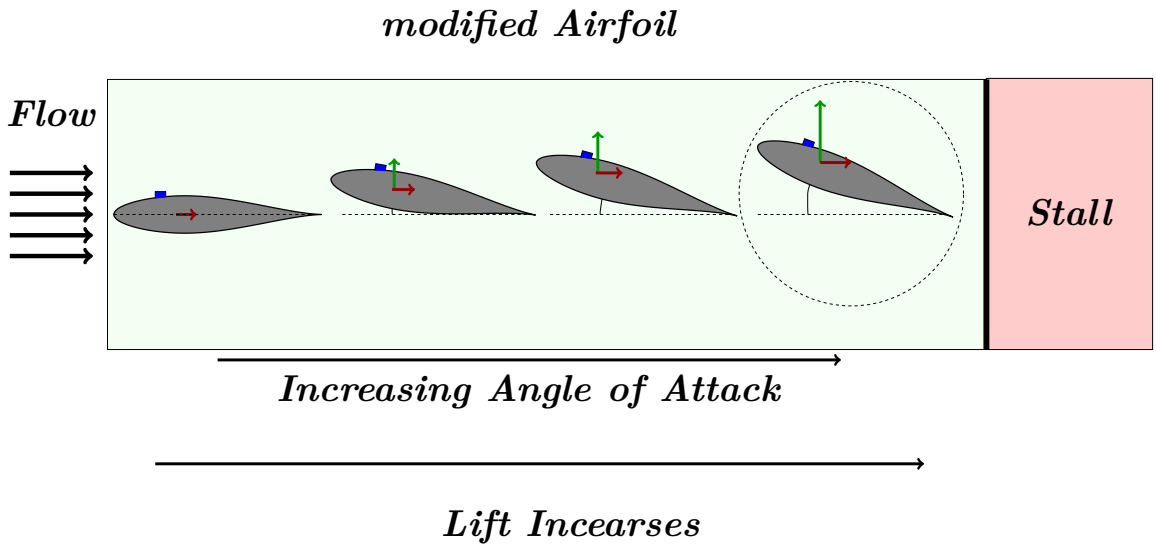


Figure 2.2: Occurrence of stall in higher angle of attacks in modified airfoils

2.3 NACA airfoils

NACA airfoils were first developed by National Advisory Committee for Aeronautics (NACA). The geometrical parameters of NACA's airfoils is depicted in figure 2.3.

NACA airfoils is comprised of many subgroup of airfoils which are : 1- Four-digit series, 2- Five-digit series, 3-Camber line profiles, 4- modified series 5- 1-series airfoils, 6- 6-series airfoils, 7- 7-series airfoils and 8- 8-series airfoils. This study will be mainly performed on NACA $63_3 - 018$ which is one of the 6-series airfoils. The next section will be presented on four-digit series since four-digit series are the most basic airfoils, and as modified series are based on the this series, having some knowledge of this serie is vital for understanding the other series.

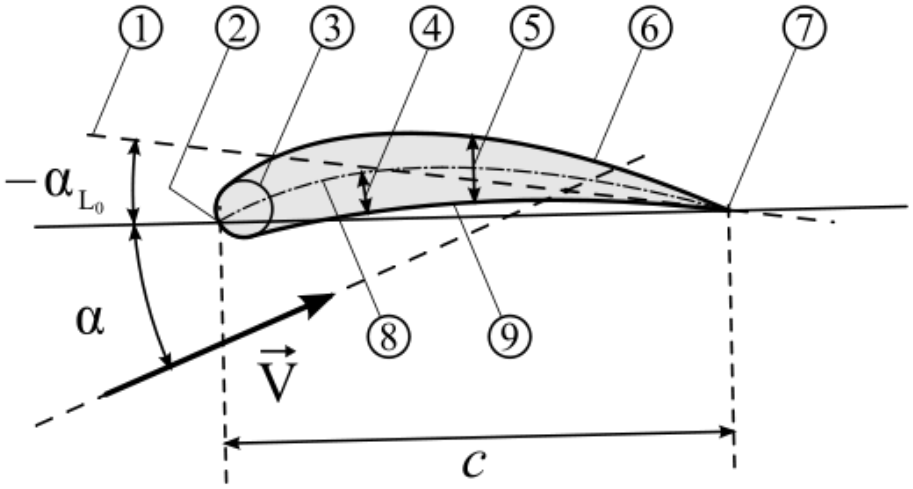


Figure 2.3: Schematic diagram of a NACA airfoil [23], 1-Zero lift line; 2-Leading edge; 3-Nose circle; 4-Camber; 5-Max thickness- 6-Upper surface; 7-Trailing edge; 8-Camber mean-line; 9-Lower surface;

2.3.1 Four-digit series

In Four-digit series, the first digit describes the maximum camber as percentage of the chord. The second digit describes the distance of maximum camber from the airfoil leading edge in tens of percents of the chord. The last two digits describe the maximum thickness of the airfoil as percent of the chord. For instance, the NACA 3518 airfoil has a maximum camber of 3% located 50% (0.5 chords) from the leading edge with a maximum thickness of 18% of the chord. In four-digit series the maximum thickness is by default placed at at 30% of the chord (0.3 chords) from the leading edge[17].

2.3.2 6-series airfoils and NACA $63_3 - 018$

"1-Series" airfoils are modified in order to maximizing the laminar flow. The six digits which describe airfoils of this series are as follow[22]:

1- The number "6" which indicates the series.

2- The next digit which describes the distance of the minimum pressure area in tens of percent of chord.

3- The subscript digit represents the range of lift coefficient in tenths above and below the design lift coefficient in which favorable pressure gradient exist on both surfaces.

4- The hyphen.

5- The digit which describes the design lift coefficient in tenths.

6- The last two digits which describe the maximum thickness as percent of chord. For the Airfoil $63_3 - 018$ has the area of minimum pressure 30% of the chord back, maintains low drag of 0.3 above and below the lift coefficient of 0, and has a maximum thickness of 15% of the chord.

Chapter 3

Literature Review

3.1 Vortex Generators

Vortex generators introduced by Taylor in 1947 [37] are known as passive devices to control the flow. These devices consist of triangular-shaped or rectangular-shaped obstacles on the suction side of the blades and could be permanently mounted on the surface or be stowed within the wing when not needed [20]. There are two mechanisms which explain why VGs enhance the performance of airfoils. First, VGs transfer the low-energy flow from the surface into the mainstream and bring the flow with high energy from the mainstream into the boundary layer where higher kinetic energy level is able to withstand a greater Adverse Pressure Gradient (APG) [11]. Second, it is suggested that excitation of local instability waves which leads to an early transition to turbulent enhance the performance of airfoils [33].

Lin [20] conducted a comprehensive review of the low profile VGs studies which includes basic fluid dynamics studies over the effectiveness of flow separation control devices at both low speed and supersonic flows, device parameters and characteristics of resulting vortices. The applications reviewed by Lin includes aerodynamics applications of increasing the lift-to-drag ratio of the low-Reynolds number airfoil, high-lift airfoils, highly swept wings, and a transonic airfoils.

The study conducted by Rao and Kariya indicates that submerged VG, which are very well immersed within the boundary layer, with $h/\delta < 0.625$ are more effective than conventional vane-type VGs ($h/\delta \sim 1$) since they result in less parasitic viscous

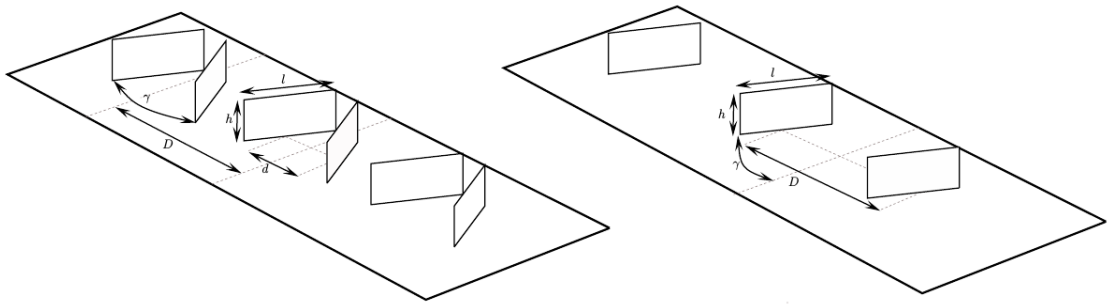


Figure 3.1: Counter rotating(left) and Co-rotating (right) VVGs

drag force [31], which at the end leads to enhancement in cruise performance [19]. Pearcey [29] studied the different types of geometrical configuration of vane-type vortex generators (VVGs). These configurations could be summarized as co-rotating, counter-rotating, multiple row of VVGs and different geometries of VVGs. Co-rotating arrangement of VVGs is usually used in the three-dimensional flow with strong cross flow. On the other hand, in counter-rotating setup, vortices with opposite rotational direction are created due to configuration of the VGs. This arrangement of VGs is especially suitable for two dimensional flows where the spanwise component of velocity is zero. These arrangements are shown in figure 3.1. Pearcey also investigated the effect of other parameters such as D, d and net length of VVGs.

Godard and Stanislas [14] investigated the parameters for an optimum configuration of a VVG. They found that the skin friction is augmented by increasing the height of the VVGs. Their results also indicate that counter-rotating configuration is two times more effective than the co-rotating configuration. They also reported that the optimum value of yaw (skew) angle,γ, is around 18°. In addition, they suggested a minimum ratio of 2 and the optimum of 2.5 for l/h.

Lin et al. [19] investigated several configurations of VGs for both flow physics and for controlling two-dimensional turbulent flow over a backward-facing ramp. They

concluded that the submerged VGs with $0.1 < h/\delta < 0.2$ placed at $5h$ to $10h$ upstream of the separation line effectively control the turbulent boundary layer.

3.2 Leading Edge Tubercle

The second passive method used in the present study is undulating leading edge of an airfoil, which is inspired by the high maneuverability of the humpback whale [10]. The idea behind this method came from tubercles on the leading edges of humpback whale flippers, as is shown in figure 3.2, which enhance maneuverability during prey capture.

The study by Watts and Fish[42] using a numerical 3D panel method showed 4.8%

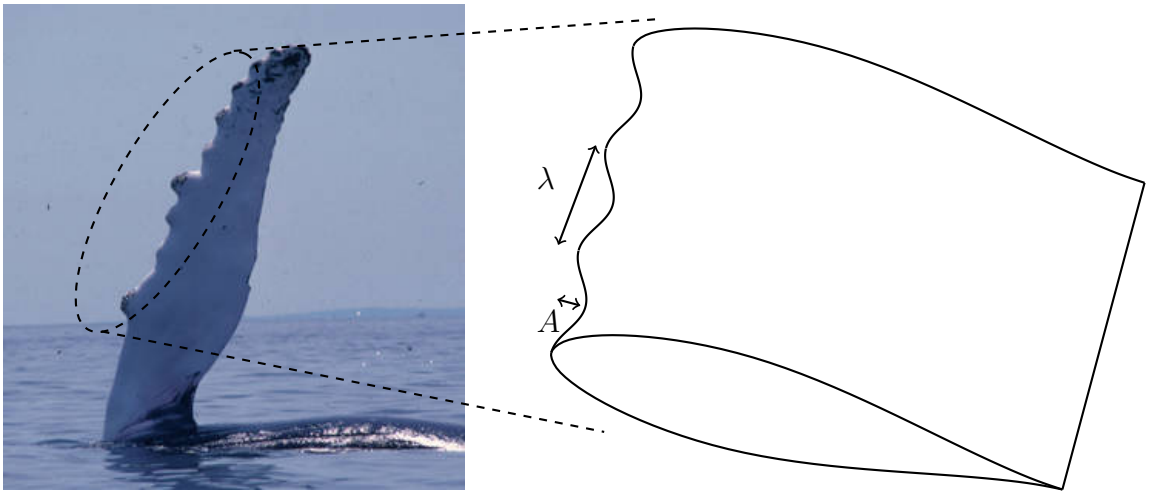


Figure 3.2: Undulating leading edge of an airfoil inspired by humpback whale

increase in lift coefficient, 10.9% reduction in drag and 17.6% increase in lift-to-drag ratio at $\alpha = 10°$. It is reported in their study that tubercles increase the performance of the wing at modest angles of attack while offering no detrimental effects at $\alpha = 0°$. Experimental study conducted by Hansen et al. [16] showed that sinusoidal undulation

over the leading edge of NACA65-021 and NACA0021 is more beneficial for NACA65-021 in comparison to their unmodified equivalent. Their results also show that reducing the tubercle amplitude leads to a higher maximum lift coefficient and larger stall angle. Nonetheless, in post-stall regime, larger-amplitude tubercles is stated to be more favorable. In addition, it is reported that reduction in wavelength leads to improvements in maximum lift and stall angle of attack. Further reduction is reported to have negative effect on performance after certain wavelength.

Flow control using spanwise sinusoidal function model is performed on a NACA0020 wing profile in a deep stall configuration $\alpha = 20°$ by Favier et al. [7]. It is reported in their study that a particular set of values for wavelength and amplitude completely changes the topology of the separated zone. In addition, it is stated in their study that streamwise vortices generated from the sides of the leading edge tubercles dominate the separated zone. For $\lambda \approx c$ and $A/c = 7\%$ configuration, it is reported that the shedding regime disappears, "streamwise vortices" generated by the leading edge tubercles dominate the flow and the boundary layer is partially attached to the wall corresponding to the pick of the tubercles.

Miklosovic et al. [25] using wind tunnel measurements showed that addition of leading-edge tubercles to NACA0020 results in 40% increase in stall angle, increase in lift and decrease in drag coefficient.

Another numerical (LES) similar research by Lin et al. [21] modified NACA0012 airfoils with 2 different wavy surfaces. The first one had sinusoidal waviness on upper and lower surfaces with a constant chord length while the second one had sinusoidal variation in both the leading and the trailing edges as well as on the upper and lower surfaces along the spanwise direction. For angle of attacks less than stall angle of attack of NACA0012 they reported a slight decrease in lift coefficient. However, for angle of attack greater than stall angle of attack of NACA0012 they observed 20%

increase in lift coefficient in the second airfoil in comparison with NACA0012.

On the other hand, the study by Stein and Murray [35] reveals that applying sinusoidal leading edge tubercles equivalent to those that are found in humpback whale increases the drag coefficient and reduces the lift coefficient. Another study by Murray et al. [26] shows that applying leading edge tubercles at different sweep angles increases the aerodynamics performance of the foil in all the sweep angles used in their study.

The underlying physical mechanisms and the associated fluid flow features are still unclear. However, it is suggested that undulation over the leading edge of airfoils acts in a similar way as vortex generators does [7]. Addition of the leading-edge tubercles is also suggested to have the same effect as vortex generators which energize the flow over an aircraft wing have. It is also speculated that the tubercles of the humpback whale are analogous to strakes used on aircraft [25].

3.3 Computation Fluid Dynamics

Among all the different approaches exist to investigate a physical phenomena, Computation Fluid Dynamics (CFD) offers a unique opportunity over experimental and/or analytical approaches. CFD can be used to model a large number of problems in a reduced cost when compared with the experimental approach. Moreover, CFD is highly desirable when analytical solution does not exist.

CFD modeling of vortex generators are mainly comprised of three approaches in the literature of the subject. These techniques are fully resolved, partly resolved and fully modeled. The first method (body fitted method) which completely includes the vortex generators in the computational domain leads to a very fine mesh close to the wall and close to the vortex generators and hence computational cost of this method is

very high in comparison to the other two techniques. In the partly modeled approach, the presence of the VG is included in the equations through the body force terms. While the mesh density required in this approach is tolerable in comparison to fully resolved method, this approach is still costly since it has to resolve the vortex structure. Finally, in fully modeled approach, even the vortex structure generated because of the presence of the vortex generators is modeled. It is worth mentioning that due to the increased modeling, the last method needs to be validated using experimental results or using results of the fully resolved method. More details about the last method could be found in the research conducted in [40, 36, 41]. Immersed Body method (IB) [30] is one of the techniques to partly resolve the flow. The important feature of this technique, which makes it a reliable candidate, is that the mesh generation in this approach is much easier than fully resolved approach. Nevertheless, although this technique needs less computational grid and thus imposes less computational time, it is still considered costly in comparison to fully modeled approach. In this research the first approach is chosen to resolve the flow as it ensures accuracy of the results.

Chapter 4

Computational method

The numerical simulation of the flow is carried out using OpenFOAM version 2.2.0. Parametric structured mesh is created in blockMesh, the OpenFoam default meshing tool. Finite Volume method is used to discretized the Navire-Stokes equations. A high Re number of 3×10^6 is chosen to benchmark the numerical model as there exist experimental results at this Re number for NACA 63-018 airfoil. A Low Re number of 5×10^5, as encountered in heavily loaded blades in hydro-kinetic vertical axis turbines, is chosen for the rest of the simulations. The $k - \omega SST$ is used with wall functions for high Reynolds simulations ($Re = 3 \times 10^6$), and without wall functions in low Re mode for low Reynolds simulations ($Re = 5 \times 10^5$). simpleFoam, a steady-state solver for incompressible and turbulent flow, is used for the range of angle of attacks from $\alpha = 0°$ to $\alpha = 18°$. If one insists on using simpleFoam in higher angle of attacks, evolution of C_l and C_d with respect to iterations shows a sinusoidal pattern as shown in figure 4.1.

pimpleFoam, a transient solver, is used in order to demonstrate the time evolution of flow flied for some of the cases. A comparison can made between the results of the unsteady solver and the steady solver by fixing every elements of the CFD model described in figure 4.1, and changing the steady solver to the unsteady solver. Results of C_l and C_d with respect to time are demonstrated in figure 4.2 when pimpleFoam is applied.

It can be seen from figures 4.1 and 4.2 that the result of lift-to-drag ratio are comparable between the two methods.

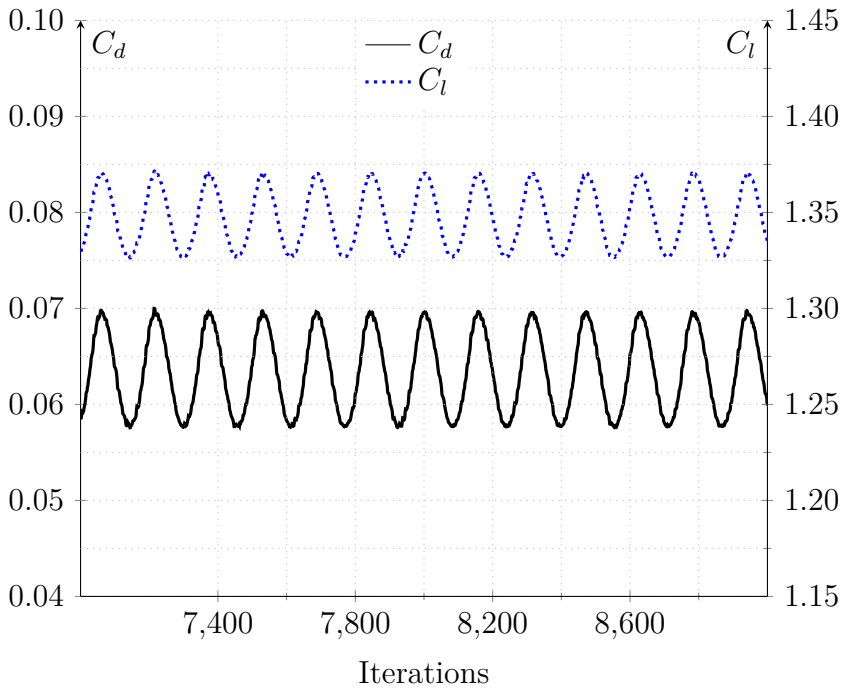


Figure 4.1: Fluctuation in the lift and the drag coefficients VS. iterations for $\alpha = 16^\circ$ and $Re = 3 \times 10^6$ using simpleFoam.

gnuplot scripts are used to monitor the lift and the drag coefficients during the simulations. Post-processing is carried out using ParaView and visit.

4.1 Meshing Technique

4.1.1 2D Mesh

Multi-block approach is used to mesh the computational domain. For a simple 2D airfoil this is carried out using 28 blocks. The schematic of the upper half of this domain is shown in figure 4.3. As it is shown in figure 4.3, the computational domain is comprised of an inner C-grid and outer O-grid and is created using blockMesh dictionary along with C++ language and m4 GNU preprocessor. C++ language is used in order to create the airfoil geometry as well as the outer O-grid, inner C-grid

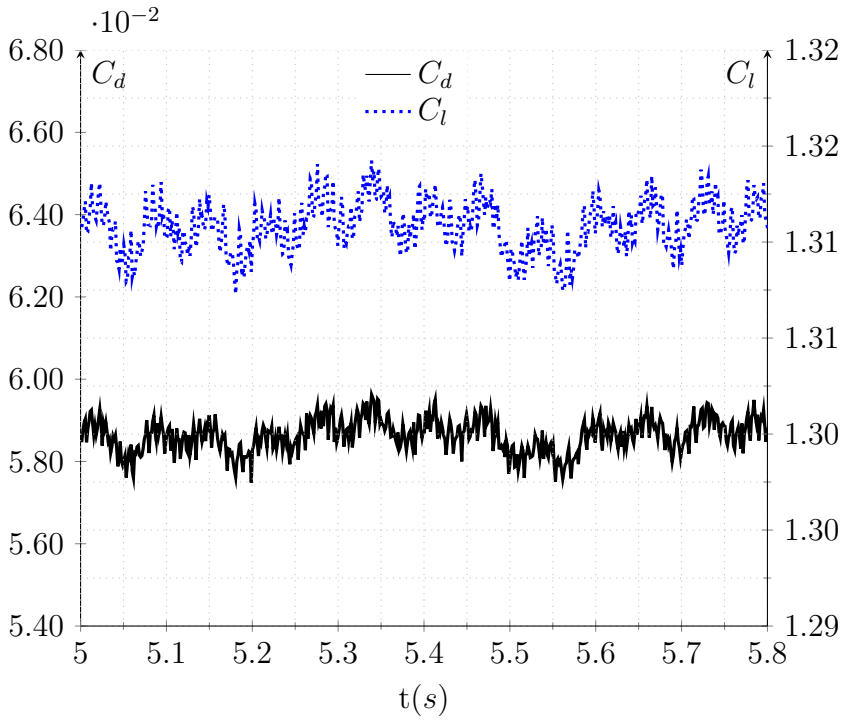


Figure 4.2: Time history of the lift and the drag coefficients at $\alpha = 16°$ and $Re = 3\times10^6$ using pimpleFoam

.

blocks and the physical size of each block as shown in figure 4.3.

In this figure $\boldsymbol{n}$ is the number of cells along each side and $\boldsymbol{l}$ specifies the physical size of each side. Using this approach allows automated meshing for any airfoil shape. m4 GNU preprocessor is used to convert a .m4 file, which contains the mesh grading and number of cells along each side, to a blockMesh file so that meshes with different resolutions and enhanced quality can be created. The quality and the resolution of the mesh and y^+ could be controlled by changing the number of cells and their grading over different edges. The resulting mesh for the 2D airfoil is shown in figure 4.4. The zoomed view of the mesh around the airfoil is shown in figure 4.5.

It is worth mentioning here that creating this mesh using fewer blocks is possible, yet in order to control the y^+ and the grading of the cells in the wall normal direction

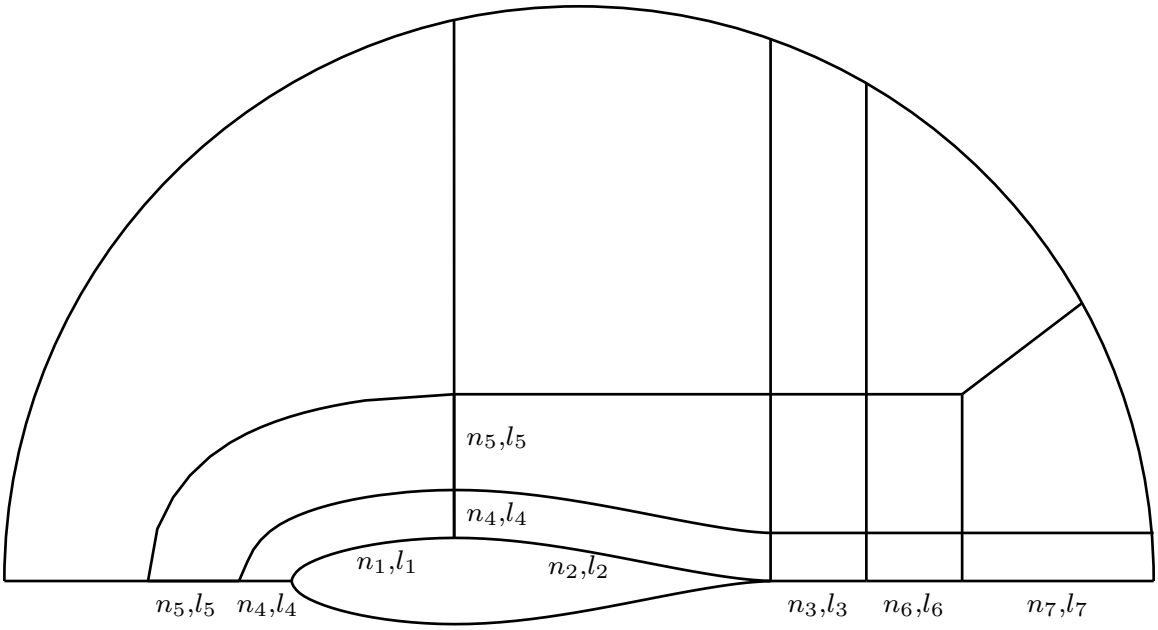


Figure 4.3: Schematic of the outer O-grid and the inner C-grid blocks and grid spacing parameters used in table 5.1 with $l_3 = c$, $l_4 = 0.3c$, $l_5 = 4.7c$, $l_6 = 9c$ and $l_7 = 90c$.

(n_4 and n_5 in the figure 4.3) in the vicinity of the airfoil, additional blocks are created to specifically perform this task while maintaining high quality cells in terms of aspect ratio of the cells far from the airfoil.

4.1.2 3D Mesh with VG

The meshing technique for 3D simulations are essentially the same as the technique used in 2D simulations. However, in the presence of the VGs, when the problem has to be solved in 3D, additional blocks should be created in the spanwise direction. Using this strategy, 198 blocks are created for simulations in blockMesh. Again, although fewer blocks could have been used to mesh the entire 3D domain, in order to have more control over the mesh of different regions, specially in the vicinity of the VG and the airfoil, and to create meshes with different resolution and enhanced quality this

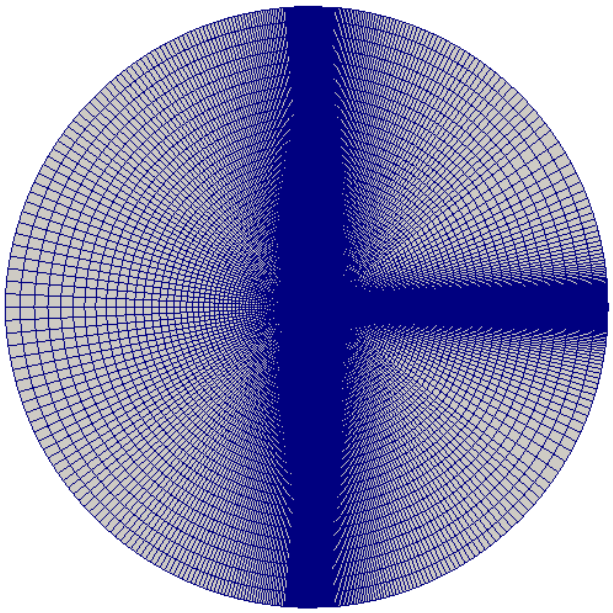


Figure 4.4: Structured mesh of 2D airfoil

strategy is chosen. The created parametric structured mesh provides the opportunity of automatically creating a structured mesh and manipulating different parameters such as the profile of the airfoil, size of the domain, place and the geometry of the VG by changing geometrical parameters. Finally, the general strategy of the meshing is to create a coarse mesh at the beginning using the algorithm discussed above and then using snappyHexeMesh to refine the mesh in the vicinity of the airfoil and the VG. To this aim, only the castellatedMesh option in the snappyHexeMesh dictionary is turned on while snap and addLayers options are off. The resulting computational domain is shown in figure 4.6.

4.1.3 3D Mesh with Undulating Leading Edge

Meshing technique for 3D airfoil with leading edge tubercles is quite the same as meshing technique used in 2D simulations. The only difference is that unlike the 2D

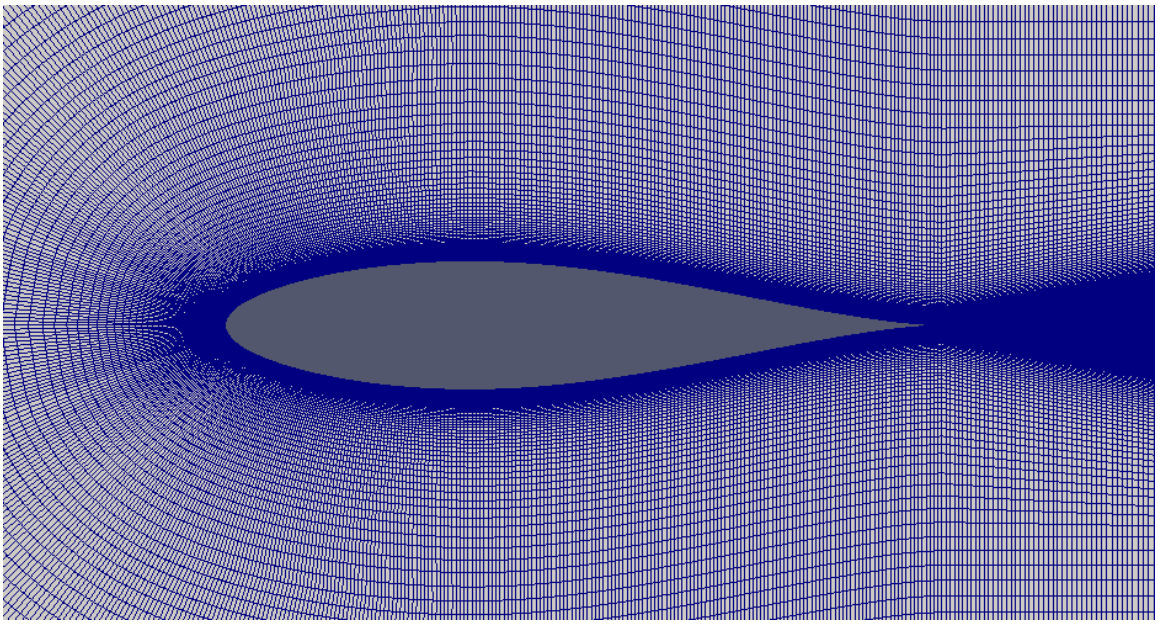


Figure 4.5: Zoomed view of structured mesh around the NACA6301 airfoil

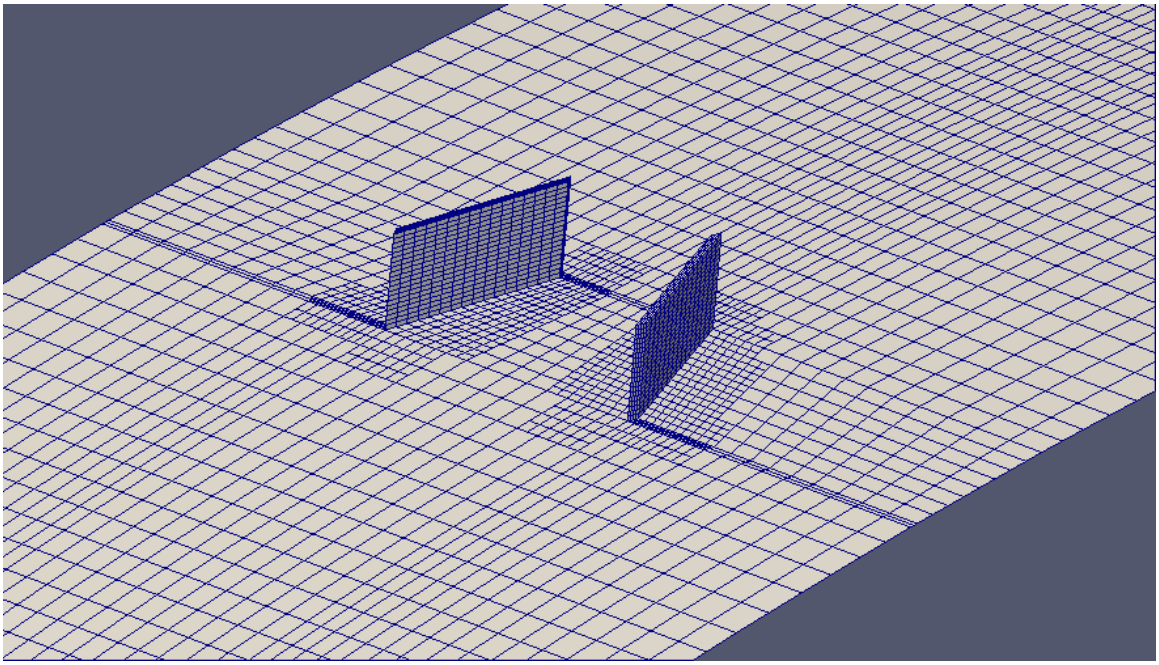


Figure 4.6: Zoomed view of the surface mesh around the VG

mesh which has only 1 cell in the spanwise direction, the 3D mesh has to have more cells in the spanwise direction so that the curvature of the sinusoidal leading edge can

be well modeled. The profile of the wavy leading edge is imported under the edges in blockMesh dictionary. The resulting computational domain is shown in figure 4.7.

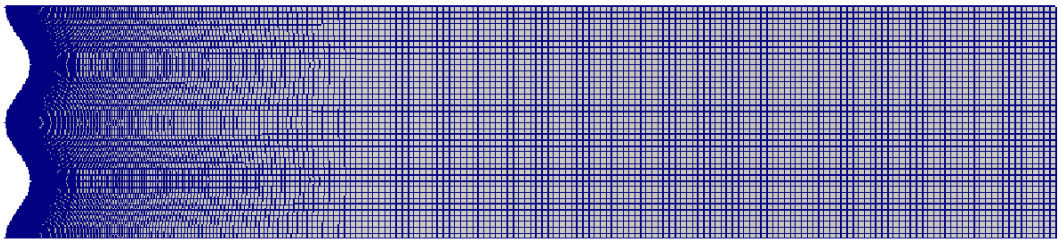


Figure 4.7: Surface Mesh of 3D airfoil with leading edge tubercle

4.1.4 Mesh Quality

Quality of the computation domain directly affects both the accuracy and the rate of convergence of a CFD problem. The higher the quality, the better the solution will be. There mainly exist five types of 3 dimensional cells that could be used in computational domains. Some of these cells are shown in figure 4.8. As discussed earlier, blockMesh and snappyHexMesh are used to create hexahedral and hexahedral-dominant meshes respectively. Hexahedron cells are chosen since the resulting computational mesh offers higher convergence speed and higher accuracy. Moreover, hexahedral meshes offer lower computational cost in comparison to tetrahedral meshes for the same accuracy inasmuch as the geometry can be meshed using fewer cells using a hexahedral mesh.

Regardless of the type of the mesh that is being used in a CFD model, quality of each mesh is measured using several criteria including aspect ratio, non-orthogonality and skewness.

Face aspect ratio of each cell is defined as the ratio of the longest length to the shortest one. Cell aspect ratio, AR, definition in OpenFOAM is the ratio of the biggest area

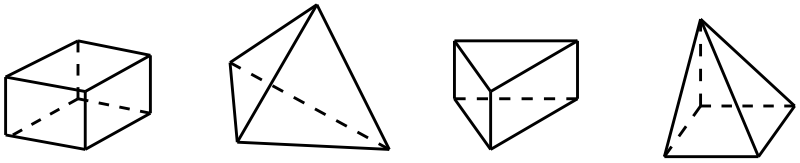


Figure 4.8: Hexahedral, tetrahedral, prism and pyramid schematics

to the smallest area of the cell's bounding box, or the following expression, the one that is larger.

$$\frac{1}{6} \times \frac{A_1 + A_2 + A_3}{v^{2/3}} \tag{4.1}$$

In equation 4.1, A_1, A_2 and A_3 are the areas of the cell's bounding box and v is the volume of the cell. $AR = 1$ is preferred whenever possible. However, CFD packages such as OpenFOAM are able to work with higher aspect ratios. In condition that the fluid flow does not have a significant component in the direction of the shortest length of the cells, higher aspect ratios are feasible. The maximum aspect ratio for the created mesh which is reported by checkMesh utility in OpenFOAM is around 600 depending on the case. This high aspect ratio is reported for the cells around the O-Grid face where the 3D dimensional effect of VGs or leading edge tubercles in the spancwise direction have been disappeared. Placing the O-Grid $100c$ away from the airfoil makes this assumption valid.

Non-orthogonality of each face is defined as the angle between the line connecting the center of the two neighbor cells of the face and the normal direction of the face. The closest the angle to zero, the better the mesh will be. However, when working with non-orthogonal meshes, in order to decrease the effect of non-orthogonal cells on the accuracy of the solution, extra pressure correction can be used in unsteady simulations to ensure that the solution is accurate at each time step. Nonetheless, extra pressure correction loops are not necessary in steady-state simulations as it is not important to have an accurate solution while the final solution is not reached. It is worth

mentioning that in OpenFOAM, both orthogonal and non-orthogonal discretizations are second order accurate. The difference is that for orthogonal cells, only the first neighbors of the cell are taken into the discretized terms, but for non-orthogonal cells, second neighbors of the cell are taken into the discretized term. Thus, working with non-orthogonal cells increases the leading terms in Taylor series and therefore increase the error [18]. In order to take the non-orthogonality of the cells into account, non-orthogonality correction option is activated using the nNonOrthogonalCorrectors under fvSolution dictionary. The maximum non-orthogonality for the meshes used in the present study and reported by the checkMesh utility is around 45° for 3D meshes consist of VGs. This high non-orthogonality is attributed to only less than 2% of total cells. Hence, considering the number of non-orthogonal cells in comparison to total number of cells and activation of non-orthogonality correction, it can be safely concluded that the resulting non-orthogonality error is negligible in this study.

Skewness is one of the other parameters which affects the CFD model. The skewness definition in OpenFOAM is though different with the definition of skewness in other CFD packages. Skewness is defined based on the ratio of m/d in figure 4.9 [18]. The

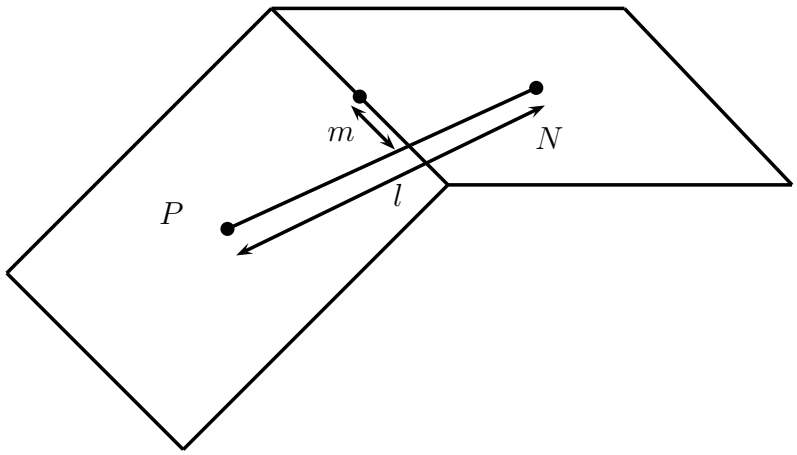


Figure 4.9: Skewness definition in OpenFOAM

maximum skewness reported by the checkMesh dictionary for the created mesh is around 1.2 depending on the case. In order to take the effect of skewed cells into account leastSquare can be used for gradient discretization in fvScheme dictionary. This will increase the computational costs but ensure the second order accuracy of the solution.

It is worth mentioning that the created mesh passes all the OpenFOAM's checkMesh criteria.

4.2 Governing equations

Numerical simulation of the problem is carried out using simpleFoam and pimpleFoam solvers. simpleFoam is a steady-state solver for incompressible and turbulent flow. simpleFoam converges fast in comparison to pimpleFoam. pimpleFoam is a large time-step transient (merged SIMPLE and PISO algorithm) solver for incompressible and turbulent flow. Using simpleFoam for high angle of attacks results in sinusoidal fluctuation in C_d and C_l graphs as it is shown in figure 4.1. Existence of these fluctuations in higher angles of attack is natural as the flow is pseudo-steady due to the vortex shedding from the trailing edge.

For the cases where pimpleFoam is used in order to present visualization of the time evolution of the flow field, $\mathbf{Co} < 0.4$ is met. Without this constrain, capturing vortex shedding from the trainling edge and obtaining a regular pattern in C_l and C_d vs. time is not possible although the simulation is stable with $\mathbf{Co} < 1$. It is argued in reference [2] that larger time steps, meaning larger **Co** numbers, can be achieved with the cost of performing additional inner iteration loops. Given the available computational resources for the present research, this is feasible for 2D simulation and **Co** can be increased up to $\mathbf{Co} = 8$, indeed. Nevertheless, for 3D simulations, the disadvantage of

this method outweighs the advantage. Thus, $\mathbf{Co} < 0.4$ with no additional inner loops is used.

The governing equations of the pimpleFoam solver are the continuity equation and the Navier -Stokes' equations of motion which are as follow:

$$\nabla.\overrightarrow{U} = 0 \tag{4.2}$$

$$\frac{\partial \overrightarrow{U}}{\partial t} + (\overrightarrow{U}.\nabla)\overrightarrow{U} = -\frac{\nabla P}{\rho} + \nu\nabla^2\overrightarrow{U} \tag{4.3}$$

Using Reynolds Averaging Navier-Stokes approach these equations result in the following equation in Cartesian coordinate:

$$\frac{\partial u_i}{\partial t} + u_j\frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial P}{\partial x_j} + \frac{\partial}{\partial x_j}[T_{ij} - \rho u_i' u_j'] \tag{4.4}$$

for an incompressible flow and using equations

$$T_{ij} = 2\mu S_{ij} \tag{4.5}$$

$$S_{ij} = \frac{1}{2}(\frac{\partial u_j}{\partial x_i} + \frac{\partial u_i}{\partial x_j}) \tag{4.6}$$

and applying the Boussinesq's assumption,

$$\rho u_i' u_j' = -2\mu_t S_{ij} + \frac{2}{3}\rho k\delta_{ij} \tag{4.7}$$

one can correlate Reynolds stress terms appeared in the equation 4.4 to k and ω using $\nu_t = \frac{a_1 k}{max(a_1\omega, \Omega F_2)}$ [24].

It is worth mentioning that ν_t defined above, rather than $\nu_t = k/\omega$, is used to modify

the original turbulent model for the adverse pressure gradient regions where the production of k is greater than its dissipation ($\Omega > a_1\omega$). Here F_2 is defined as the absolute value of the vorticity. The modified equations for k and ω are as follows:

$$\frac{D\rho k}{Dt} = -\beta^*\rho\omega k + \frac{\partial}{\partial x_j}[(\mu + \sigma_k\mu_t)\frac{\partial k}{\partial x_j}] + \tau_{ij}\frac{\partial u_j}{\partial x_j} \tag{4.8}$$

$$\frac{D\rho\omega}{Dt} = -\beta_2\rho\omega^2 + \frac{\partial}{\partial x_j}[(\mu + \sigma_\omega\mu_t)\frac{\partial \omega}{\partial x_j}] + \frac{\gamma_2}{\nu_t}\tau_{ij}\frac{\partial u_j}{\partial x_j} + 2\rho(1 - F_1)\sigma_{\omega 2}\frac{1}{w}\frac{\partial \omega}{\partial x_j}\frac{\partial k}{\partial x_j} \tag{4.9}$$

Constants of the last two equations can be found in reference [24].

Here, the lift and drag coefficients are defined as follow:

$$C_l = \frac{F_l}{\frac{1}{2}\rho U_\infty^2 A} \tag{4.10}$$

$$C_d = \frac{F_d}{\frac{1}{2}\rho U_\infty^2 A} \tag{4.11}$$

where $A = cw$ is the projected area of the airfoil and F_l and F_d are the net forces acting respectively in the normal and parallel direction with respect to the incoming flow.

4.3 Boundary Conditions

Boundary conditions of a CFD models is one of the most important factors that affects the accuracy/convergence of simulations. Choice of boundary condition is not unique; meaning that to different boundary conditions may result in the same final result as long as both are physically valid. However, non-physical boundary conditions quickly result in divergence of solution at the very beginning of the simulation. For

example, both zero flux and/or fixed value of a variable in far-field boundary are valid for domains such as the one presented in figure 4.4, yet a constant value of 0 for velocity in that surface results in divergence of the solution as it obviously violates the continuity condition. Therefore, before anything else, making sure that the boundary conditions are correct is the priority in solving a CFD problem.

Boundary conditions of the model are fixedValue of 0 and U_∞ at the airfoil wall and O-grid faces respectively and zeroGradient for pressure at both of the airfoil wall and O-grid faces . Boundary conditions of k and ω are inletOutlet at the O-grid faces for both of these parameters and kqRWallFunction and omegaWallFunction at the airfoil wall for k and ω respectively for high Reynolds ($Re = 3 \times 10^6$) simulations. For High Reynolds simulations $y^+ \approx 100$ is used. This allows for lower mesh resolution in vicinity of the wall and therefore decreases the computational time. It is worth mentioning that maintaining $y^+ < 1$ for such a high Re flow results in extremely small cells around the airfoil. Moreover, specially close to the wall, cell growth factor is limited to $1.05 - 1.1$ for accurate calculation of boundary layer. Hence, inasmuch as maintaining $y^+ < 1$ is computationally extremely expensive at high Re numbers, $y^+ \approx 100$ is used. This is achieved by changing the grading and number of cells along l_4 in figure 4.3. However, with $y^+ \approx 100$ no physics inside the viscous sublayer can be captured. Therefore, for these cases, wall functions are used to approximate the fluid flow inside the first cell. For Low Reynolds simulations ($Re = 5 \times 10^5$) maintaining $y^+ < 1$ is achievable withing the available computational resource. In these cases no wall function is used and the flow field inside the viscous sublayer is fully resolved. Therefore, a very small value for k and ω is used at the airfoil and the VG.

Boundary conditions of the front and back faces are empty for 2D simulations. This BC type allows for 2D calculation of flow flied in OpenFOAM. For 3D airfoil with

VG, symmetryPlane is used for front and back faces. However, the validity of this last assumption should be investigated since applying symmetric condition between the barriers of one pair of vortex generator could be argued because of the 3 dimensional nature of turbulence. In order to validate this assumption, a full model of a pair of VG, as shown in the figure 4.10 in the right, is compared to the one that models only half of the problem, as shown in the figure 4.10 in the left . The results of the C_d

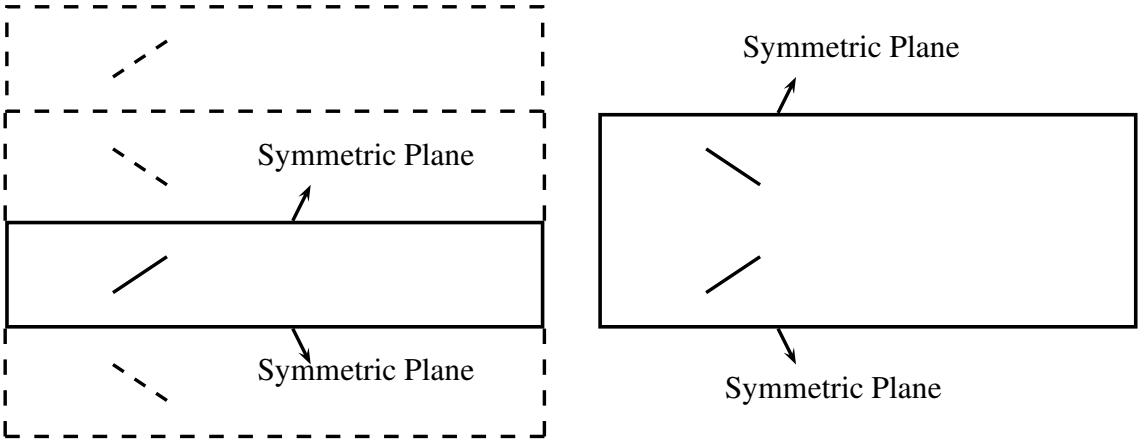


Figure 4.10: Top view schematics of the airfoil for 3D simulations with VGs. The half model (Left) and the full model (Right)

and the C_l for full model simulation are in very good agreement to the results of half model. Therefore, the half model is used for 3D airfoil with VGs.

For 3D airfoil with undulating leading edge, cyclic boundary condition is used for the Front and Back patches as is shown in figure 4.11.

Cyclic (Periodic) boundary condition allows for simulating repetitive geometries such as the case which studied in this study. A summary of the boundary conditions of the CFD model are presented in table 4.1. Here, omegaWallFunction and kqRWallFunction are the wall function used at the wall for ω and k respectively.

Mathematical definitions of the boundary conditions used in table 4.1 is presented in table 4.2. In this table, **patch** refers to one of the faces used in the first column of

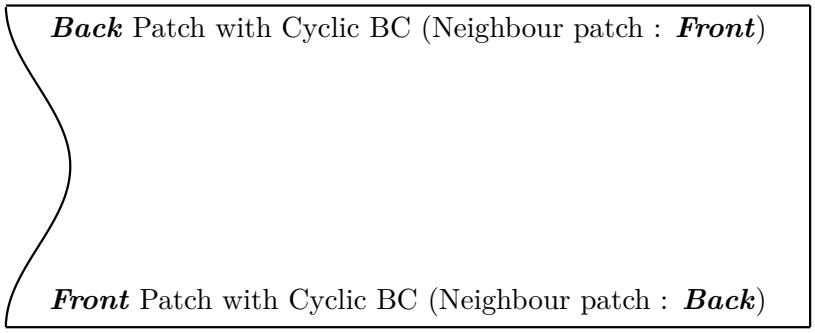


Figure 4.11: Top view schematic of the 3D airfoil with undulating leading edge

Variable / ***Patch***	U	p	k	ω
Airfoil [1]	fixedValue value= (0 0 0)	zeroGradient	kqRWallFunction value=10^{-8}	omegaWallFunction value=1350
Airfoil [2]	fixedValue value= (0 0 0)	zeroGradient	fixedValue value=10^{-8}	zeroGradient
O-Grid	fixedValue value= $\overrightarrow{U}$	zeroGradient	turbulentIntensityKineticEnergyInlet intensity=0.3	fixedValue value=1350
Front and Back for 2D airfoil	empty			
Front and Back for 3D airfoil [3]	symmetryPlane			
Front and Back for 3D airfoil [4]	cyclic			

Table 4.1: Boundary conditions of the CFD model
1)at high Re, 2)at low Re, 3)with VG and 4)with leading edge tubercle

the table 4.1, n is the normal direction to the patch and θ is one of the variables used in table 4.1.

BC type	***Mathematical definition***
fixedValue	$\theta\|_{patch} = \text{value}$
zeroGradient	$\frac{\partial\theta}{\partial n}\|_{patch} = 0$
turbulentIntensityKineticEnergyInlet	$I\|_{patch} = \text{value}$
empty	$\theta = 0$ and $\frac{\partial\theta}{\partial n} = 0$
symmetryPlane	$\frac{\partial\theta}{\partial n}\|_{patch} = 0$
cyclic	$\frac{\partial\theta}{\partial n}\|_{patch} = \frac{\partial\theta}{\partial n}\|_{neighbor}$ and $\theta\|_{patch} = \theta\|_{neighbor}$

Table 4.2: Mathematical definition of BCs used in OpenFOAM

4.4 Finite Volume Schemes

Choice of FV schemes is of great importance in every CFD problem. Purely second order schemes are not usually stable enough and bounded, depending on the quality of the mesh that is being used, although they are very accurate. On the other hand, first order schemes are bounded and stable although are not accurate. Therefore, a blend of second order schemes with added dissipation of first order schemes is used in the present research to ensure the accuracy and stability of the solution.

In the finite volume schemes, Gauss linear is used as the default gradient discretization scheme, but cellLimited Gauss linear 1 is used for grad(U). cellLimited adds some dissipation to the linear discretization and thus improve the stability of the solution. linearUpwind cellLimited Gauss linear 1 is applied to divergence terms, and Gauss linearLinearV cellLimited Gauss linear 1 is used for velocity divergence. Also bounded Gauss is used in convective schemes to maintain the bounds of strictly bounded variables such as k and ω. In addition, Gauss linear corrected 1 is used for laplacian terms. Finally, linear scheme is applied to interpolationSchemes and fourth to snGradSchemes.The

aforementioned set of schemes found to be accurate in spite of the fact that dissipative schemes are used in order to stabilize the solution.

4.5 Solution Control

Solution speed in CFD problems is mainly a strong function of linear system solvers. In the present study, for solving the resulting system of linearized equations, PBiCG is used as the solver and DILU as the preconditioner for all variables except for the pressure. For the pressure equation in serial simulations GAMG is applied. However, in parallel simulations better speedup is obtained using PCG solver and DIC preconditioner. Absolute and relative tolerances for all variables in the steady simulation are 10^{-6} and 10^{-1} respectively except for the pressure equation where 10^{-2} is used as the relative tolerance to speed up the solution. This rather large relative tolerance, unlike unsteady problems, does not affect the accuracy of the simulation in steady-state problems and the solution is reached after certain number of iterations even if the solution is not converged at each iteration. For unsteady simulations, however, the solution should be converged at each time step. Therefore, 10^{-6} for absolute and 0 for relative tolerance are used for every variable [9].

During the solution process, residuals, lift and drag coefficient are monitored. Convergence of the lift and the drag coefficients rather than residuals is used to terminate solutions. This ensures that the converged solution is obtained. otherwise, if one relies solely on residuals in order to judge the convergence of the solution it may result in an inaccurate final result. In short, stability of the solution which is obtained based on residual plot is not a measure of accuracy. Hence, other criteria is required to judge the accuracy of the solution as is discussed in the next chapter.

The plot of the residuals is shown in figure 4.12 for a test case. It can be seen that after 12,000 iterations residuals of the variables remains low and in an approximately constant range. Figures such as 4.1 and 4.2, as discussed earlier, are used to judge the convergence of solutions.

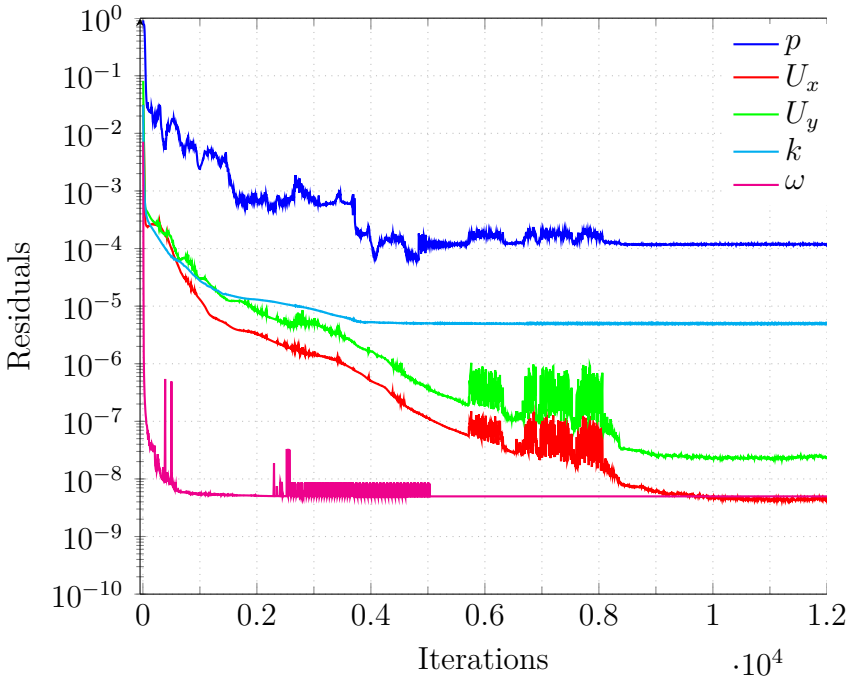


Figure 4.12: Absolute residuals of variables at $\alpha = 8^{\circ}$ and $Re = 5 \times 10^{5}$

Chapter 5

Results

5.1 Validation of numerical results

Careful grid independence study is conducted on the computation grid. The important point that should be mentioned here is about the size of the computational domain that is necessary to avoid boundary effects on the lift and the drag coefficient. It is found in the present research, that an O-grid with radius around $100c$ is required to avoid far field boundary effects on the results of the C_l and C_d. In addition, schematic of the domain that demonstrates the grid independence parameters, and the results of the grid independence study are shown in figure 4.3 and table 5.1 respectively. As it appears, results, the drag coefficient in particular, are very sensitive to the number of cells along the wall (n_1 and n_2 in the figure 4.3) up to case number 3. Increasing the number of cells in this direction does not change the results in a great extent after case number 3. Finally, Grid number 3 is selected for all the simulations as it leads to less computational cost together with desirable accuracy.

Results of the lift and the drag coefficients obtained from the simulations of 2D bare

Case	n_1	n_2	n_3	n_4	n_5	Kcell	C_l	C_d	$C_{l_{\text{error\%}}}$	$C_{d_{\text{error\%}}}$
1	30	20	20	60	60	46	1.28	0.08	14.9	92.3
2	50	50	50	60	60	110	1.41	0.057	6.3	37.0
3	75	75	75	65	60	162	1.52	0.043	1.0	3.3
4	75	75	75	105	160	275	1.51	0.041	0.3	1.4
5	100	100	100	105	160	388	1.5	0.04	0	0

Table 5.1: Grid independence study parameters

airfoil are compared to the experimental data published for NACA63018 [3]. As it can be seen from figure 5.1, when compared to experimental data, results for a bare airfoil show good agreement in the lift coefficient before the stall region while the data in the stall and post stall regions are slightly overpredicted. The drag coefficient is overestimated in low angles of attack since a RANS model is used. RANS models assume turbulent flow all over the domain which is not the case in practice. Therefore, they tend to over-predict the drag coefficient since the drag is mainly viscous driven and RANS or URANS models are not able to accurately predict the viscous sublayer. On the other hand, prediction of the lift coefficient using RANS or URANS is accurate as the lift is mainly pressure driven.

In the present study, RANS and URANS are used to resolve the turbulent flow and the results of CFD for 3D airfoils are compared with the results of CFD for 2D airfoil. Thus, although the CFD model over-predicts the drag, use of a RANS model is justified inasmuch as final result of lift-to-drag ratio is used for comparison.

Therefore, with that in mind that the CFD predictions of the C_l for the bare airfoil is slightly over-estimated in the stall region and C_d prior to the stall, results of the 3D airfoil cases consists of the VGs (passive airfoils) are compared with the results obtained using CFD for the bare airfoil.

Moreover, in order to make sure that the results of 3D case do not depend on the structure of the blocks and that the 3D domain could duplicate the results of a bare airfoil a validation test is performed. To this aim, the VG block is converted from a solid domain to a fluid domain in 3D mesh. If the 3D solution is independent of the mesh structure this test should predict the characteristics of a bare airfoil. The result obtained for 3 angles of attack validates the 3D model.

In addition, in order to make sure that the result of 3D airfoil with undulating leading edge does not dependent on the structure of the mesh a validation test is performed

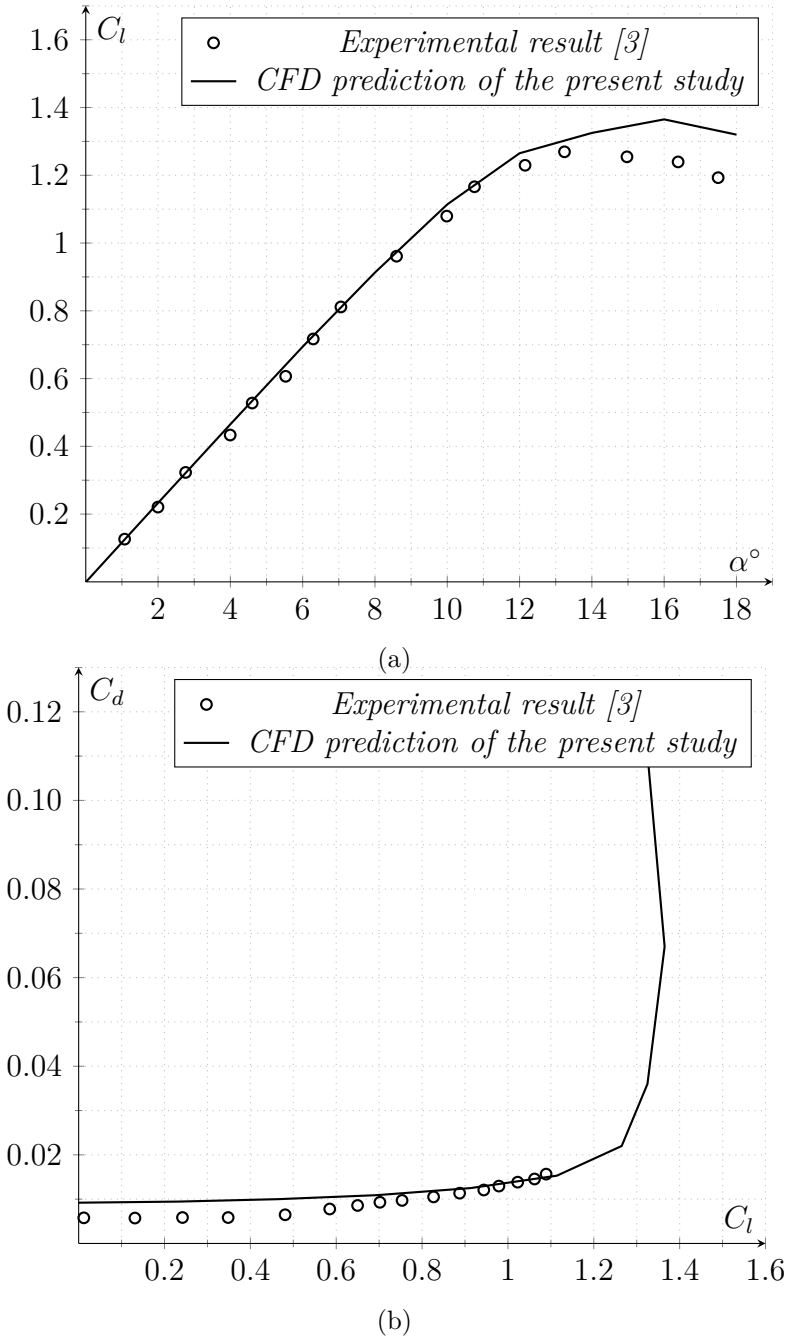


Figure 5.1: Comparison of the experimental results published in [3] and CFD prediction of the present study at $Re = 3 \times 10^6$. a) Lift Vs. α and b)Drag Vs. Lift

on the amplitude of the leading edge tubercles. Starting form a large amplitude and decreasing the amplitude results in bare airfoil characteristics.

5.2 Vortex generator

To obtain the optimized configuration of the VG for this particular airfoil several configurations are tested. The starting guesses could be found in references [20, 19]. The results of this study, which is conducted at $AOA = 16^\circ$ and $Re = 3 \times 10^6$ is shown in table 5.2. It is worth mentioning that the height of the vortex generators selected in this study is in the order of the height of vortex generators first proposed by Taylor [37]. It is found that the sub-merged vortex generators, widely investigated by Lin [19, 20], does not improve the performance of this airfoil (Case number 5 in table 5.2) as much as conventional vane type does. Finally, Case number 6 is chosen in order to investigate the effect of vortex generator on the NACA 63018 at range of angle of attacks from $\alpha = 0^\circ$ to $\alpha = 18^\circ$.

In lower Reynolds Numbers, $Re = 5 \times 10^5$ for instance, since the separation happens

Case	D/d	$\Delta x/c$	h/d	l/d	C_l	C_d
Bare airfoil	–	–	–	–	1.41	0.057
1	3.5	0.3	0.5	1	1.61	0.045
2	3.5	0.15	0.5	1	1.62	0.043
3	3.5	0.2	0.5	1	1.63	0.042
4	5	0.2	0.5	1	1.57	0.048
5	1.75	0.2	0.25	0.5	1.62	0.047
6	6	0.2	1	2	1.65	0.042

Table 5.2: Effect of the vortex generator geometrical parameters on the lift and the drag coefficients at AOA=16°

earlier due to generation and advection of the laminar separation bubble, the situation is in favor of the airfoil implemented with vortex generators. The stall begins in lower

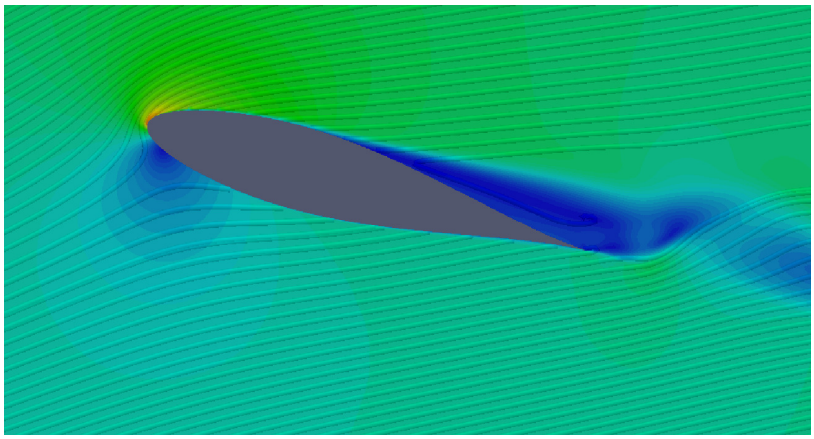
angles of attack and thus using VGs that increase inertia forces and consequently decrease the form drag is thought to benefit the airfoil even more. However, as the separation point is not a fixed point on the airfoil and is different to that of high Reynolds numbers, another design of the vortex generator in terms of D/d, $\Delta x/c$, h/d, l/d, and γ is required. In general, for lower Reynolds Numbers, the VGs have to be shifted upstream since the separation occurs close to the leading edge. At $Re = 5 \times 10^5$, similar configuration as the case 6 (table 5.2) but with $\Delta x/c = 0.1$ and $\gamma = 25$ is found to be more effective than the other configurations. In low Reynolds number as the inertia forces are less effective in resisting the adverse pressure gradient, yaw angle, γ, is increased to $\gamma = 25°$ in order to further strengthen the downstream vorticities.

Velocity contours are demonstrated in figure 5.2 for the bare airfoil and the modified airfoil. Velocity profiles indicate that the separation is prevented if VGs are applied.

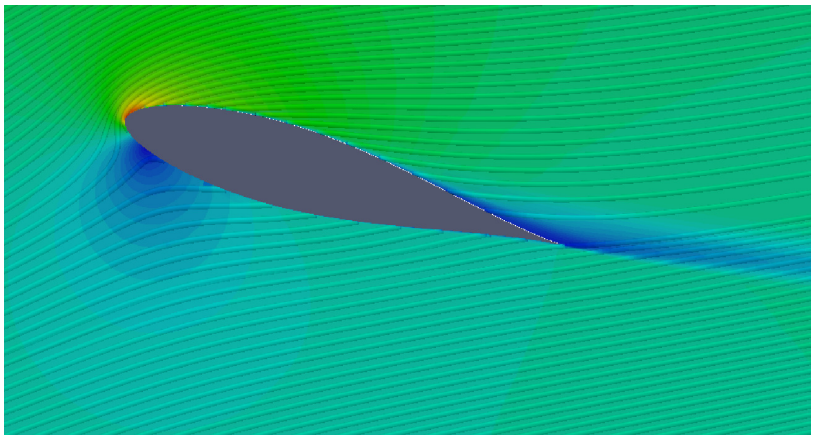
Moreover, velocity profiles at different downstream sections are shown in figure 5.3 to show how velocity profiles look like. It can be seen that because of the presence of the VGs, vortices are generated downstream(figure 5.3) and suppress the separation (figure 5.2).

Vorticity profile can be used to further investigate the effect of presence of VGs on the downstream flow. Vorticity profile are plotted in different spanwise sections in figure 5.4. As it can be seen, presence of VGs results in an attached flow in every section across the span while signs of flow separation can be seen for the bare airfoil. The physical mechanics behind this can be assumed as if the VGs act locally such as an "hydro-foil" at a local angle of attack. This angle is set using γ as is shown in figure 3.1. This causes circulation around VGs and the vortices behind the VGs energize the flow and subduce the boundary layer separation.

In addition, results of the time evolution of the vorticity profile are shown for different



(a) Bare airfoil



(b) Airfoil with VG

Figure 5.2: Comparison of the velocity profiles at $\alpha = 16°$ between the bare airfoil and the one with VG

time steps for both the bare airfoil and the modified airfoil in figures 5.5 and 5.6 respectively. Plots shows evolution of vorticity in two characteristic time scale.

It can be seen in figure 5.6 that by using VGs, vortex shedding is suppressed. The characteristic vortex shedding time scale is obtained using $t_s = c/U_\infty = 0.33$, assuming that $\nu = 10^{-6}\frac{m^2}{s}$ and $c = 1m$. Results shown in figure 5.5 show that characteristic vortex shedding time scale using CFD is in very good agreement with the above theoretical calculation. Moreover, it can be concluded that since vortex shedding is suppressed, fluctuation in the lift and drag coefficient is damped as well. This benefit

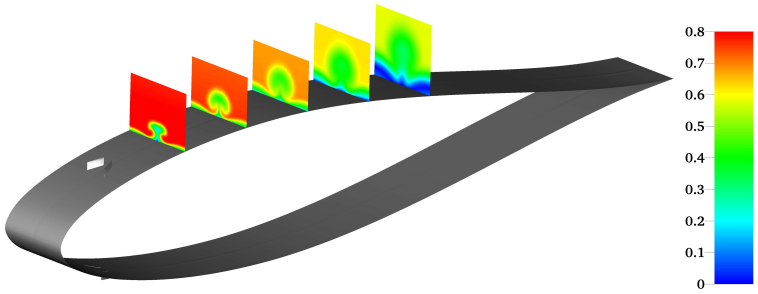


Figure 5.3: Velocity profiles (m/s) at different sections downstream the flow

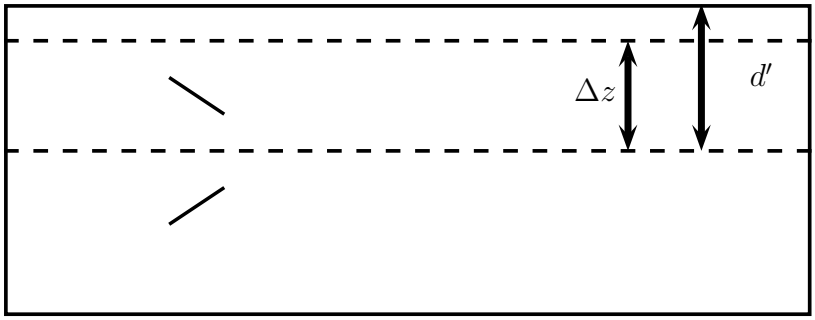
the structure of the blades as they experience less force fluctuation as a result.

As stated before, applying VGs on the blades of a vertical axis turbine requires that two pairs are placed on the pressure side and the suction side. Applying another pair of VGs will increase the drag penalty that is paid. Nevertheless, when compared with the passive airfoil that has only a single pair of vortex generator on the suction side, the average lift in one turn of rotation obtained from the passive airfoil with double sided VGs on the top and the bottom will be greater.

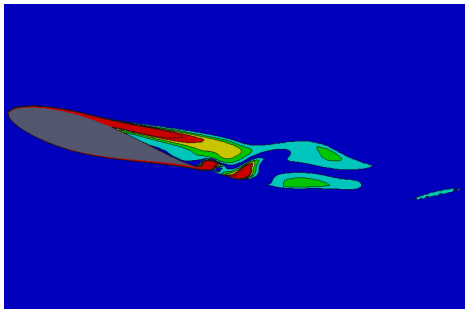
Results of the variation of the lift and drag coefficients with angle of attack for both cases at $Re = 3 \times 10^6$ are shown in figure 5.7 along with the CFD data of the bare airfoil.

It can be seen that the lift coefficient is improved in the stall region when vortex generators are used. The downside of using the single vortex generators configuration in comparison to the bare airfoil is in the pre-stall region where separation does not take place and thus, vortex generators do not improve the lift in this region although they increase the viscous drag.

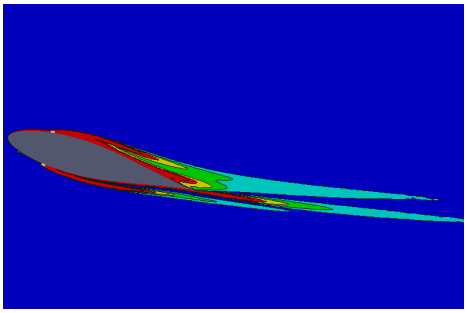
It also can be seen that compared with passive airfoil with single pair of VG on suction side, passive airfoil with double pair of VGs on suction side and the pressure side has the same C_l vs. α specification and that is because of the fact that the presence of



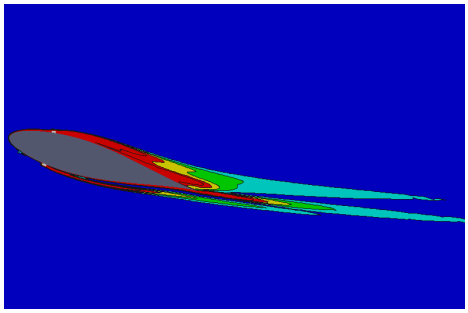
(a) Definition of Δz and d' for each section



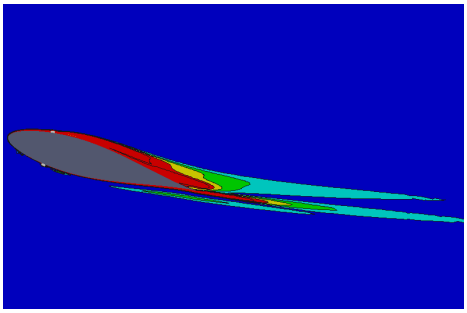
(b) Bare airfoil



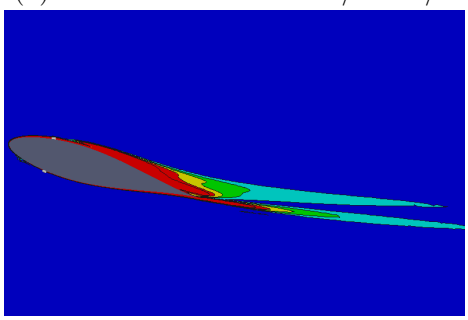
(c) Modified airfoil with $\Delta z/d' = 1/6$



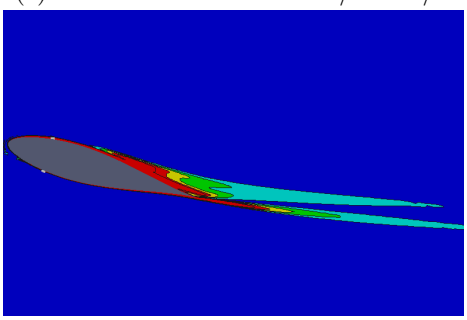
(d) Modified airfoil with $\Delta z/d' = 2/6$



(e) Modified airfoil with $\Delta z/d' = 3/6$

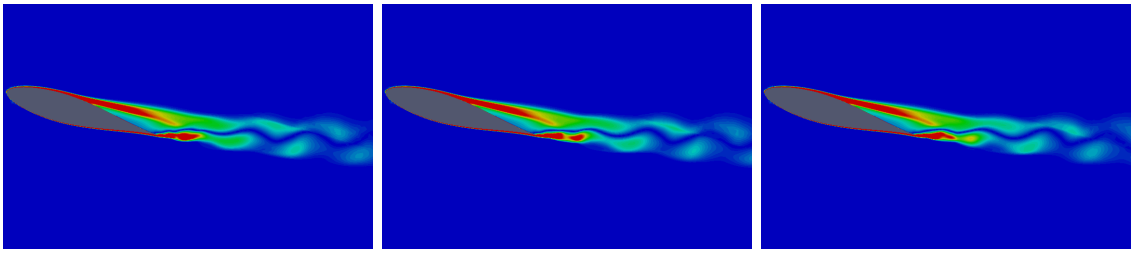


(f) Modified airfoil with $\Delta z/d' = 4/6$

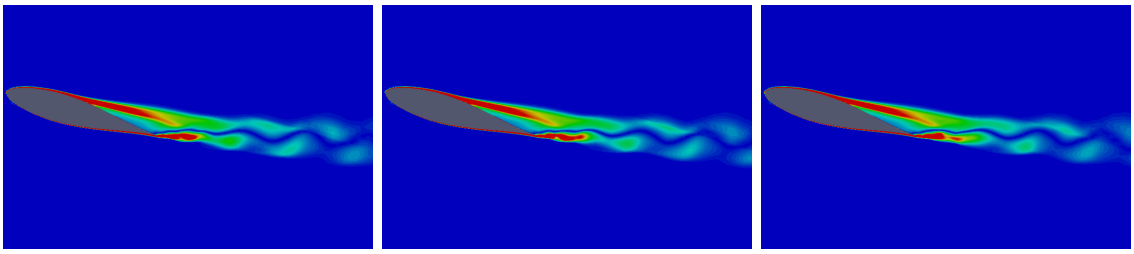


(g) Modified airfoil with $\Delta z/d' = 5/6$

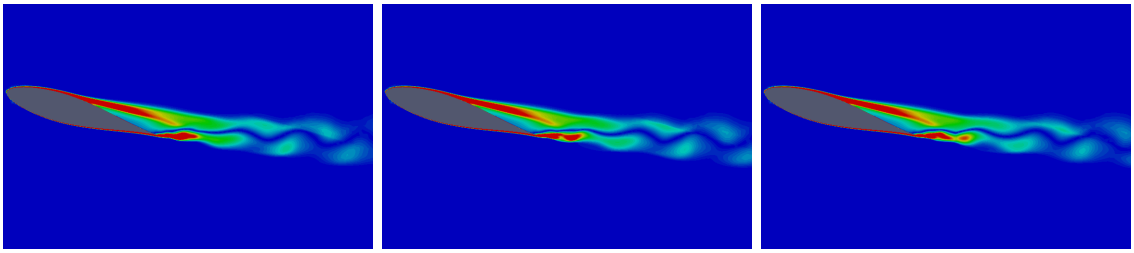
Figure 5.4: Vorticity profiles at different sections along the spanwise direction



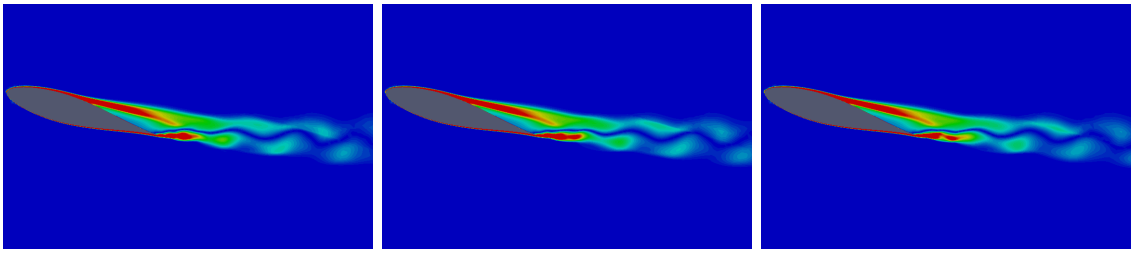
(a) $t = 0s$ (b) t=0.10s (c) t=0.15s



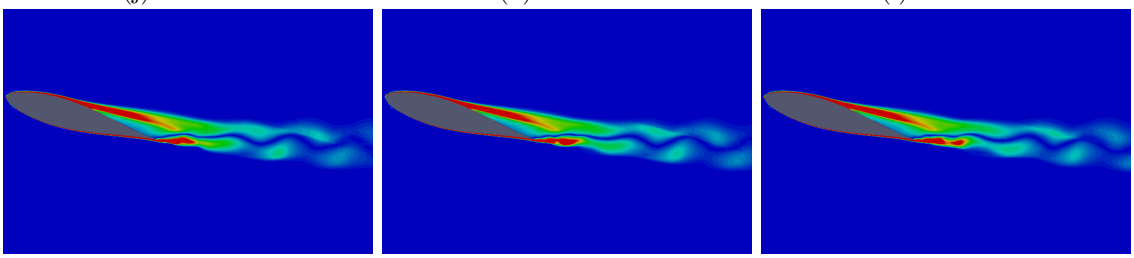
(d) t=0.20s (e) t=0.25s (f) t=0.30s



(g) t=0.35s (h) t=0.40s (i) t=0.45s

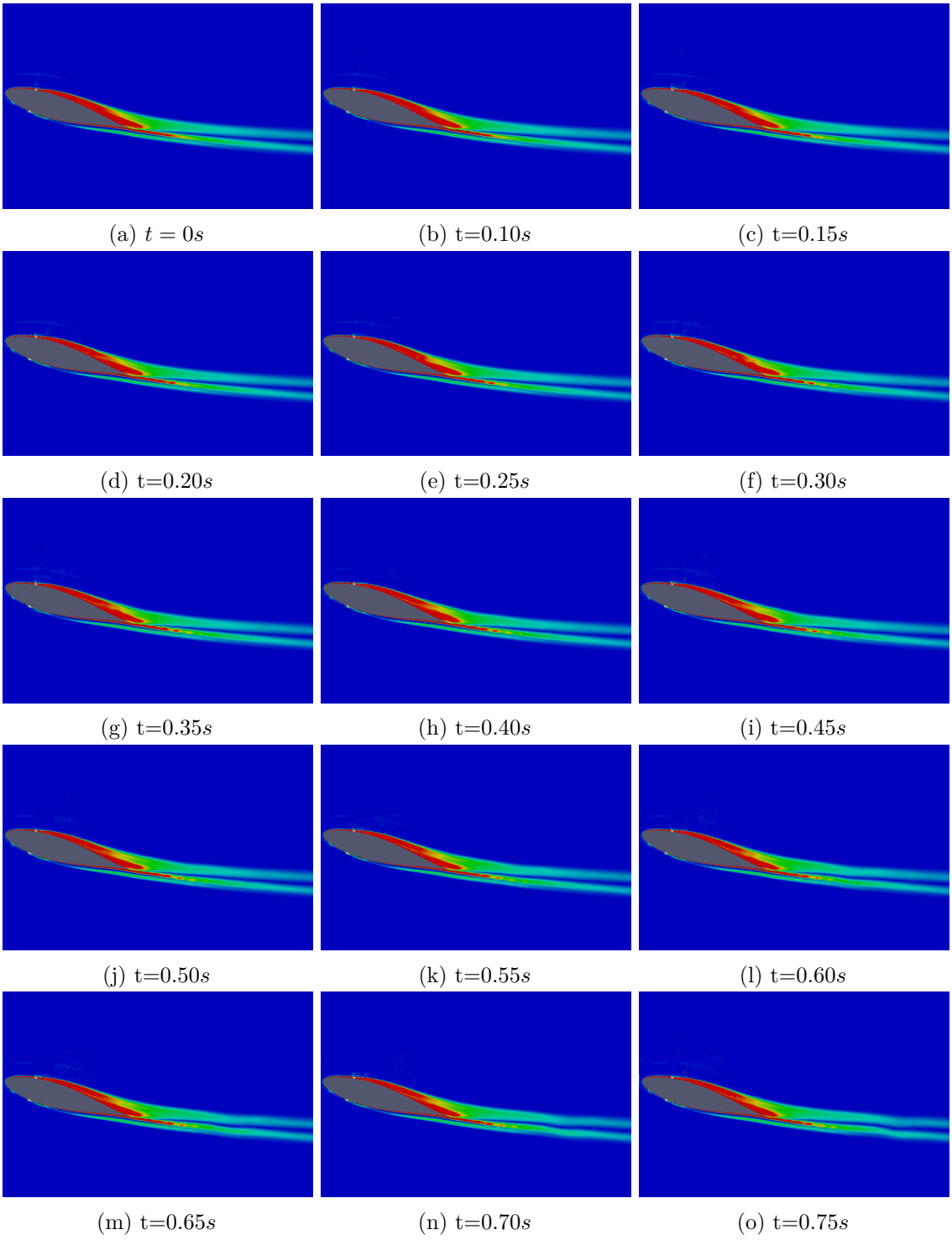


(j) t=0.50s (k) t=0.55s (l) t=0.60s



(m) t=0.65s (n) t=0.70s (o) t=0.75s

Figure 5.5: Time evolution of the vorticity profiles at $\alpha = 16^{\circ}$ and $Re = 3 \times 10^3$ for the bare airfoil with characteristic time scale $t_s = c/U_\infty = 0.33$



(a) $t = 0s$ (b) t=0.10s (c) t=0.15s

(d) t=0.20s (e) t=0.25s (f) t=0.30s

(g) t=0.35s (h) t=0.40s (i) t=0.45s

(j) t=0.50s (k) t=0.55s (l) t=0.60s

(m) t=0.65s (n) t=0.70s (o) t=0.75s

Figure 5.6: Time evolution of the vorticity profiles at $\alpha = 16°$ and $Re = 3 \times 10^3$ for the 3D airfoil with VGs. Sections at $\Delta z/d' = 4/6$

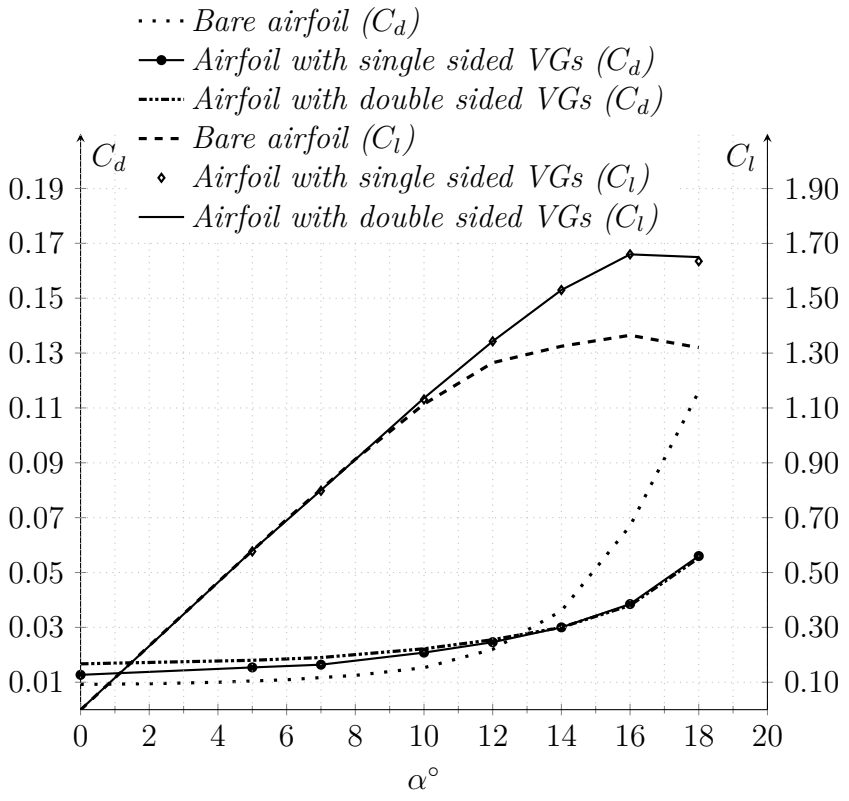


Figure 5.7: Effect of using single-sided vortex generators on top and double-sided vortex generators on top and bottom sides at $Re = 3 \times 10^6$

the VG on the pressure side does not improve the performance of the airfoil unless this side becomes the suction side after 180° of rotation.

Therefore, The downside of using the passive airfoil with double sided VGs in comparison to the passive airfoil with single sided VG is mostly in the pre-stall region. As a result of adding another vortex generators on the pressure side, although the lift coefficient does not change, because of the presence of an extra pair of VG, the viscous drag increases even more and this ultimately increases the drag coefficient. This is the penalty that one pays when using vortex generators.

The result of applying double VGs on the airfoil at low Re Number of $Re = 5 \times 10^5$ is shown in figure 5.8. This is a typical condition for small VATs that operate underwater and are the interest of the current research.

It can be seen form the figure 5.8 that at $Re = 5 \times 10^5$ the passive airfoil behaves

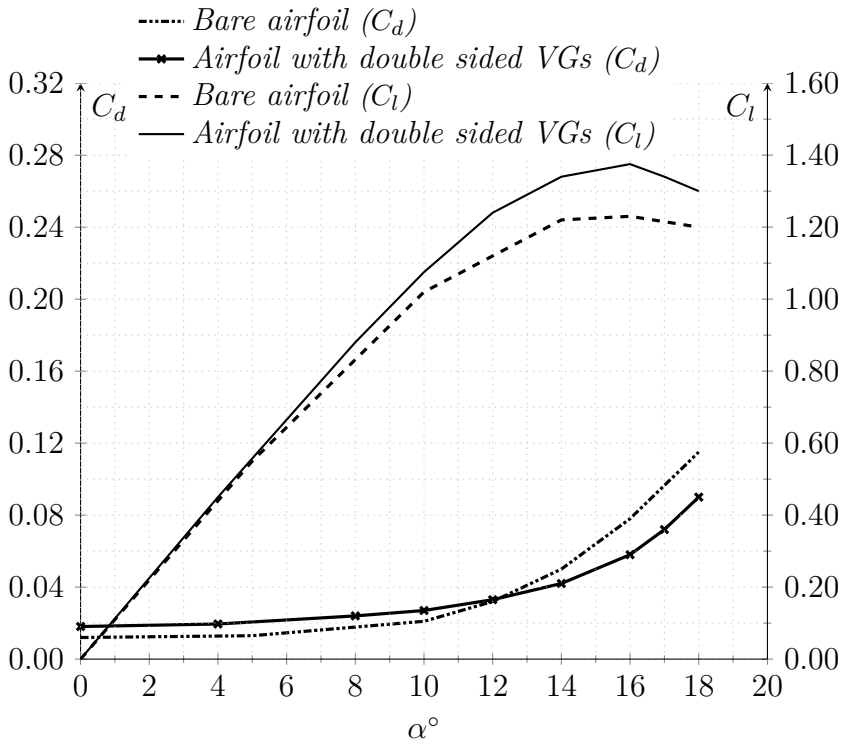


Figure 5.8: Effect of using double-sided vortex generators on the suction side and the pressure side at $Re = 5 \times 10^5$

almost in the same way as the one that works in high Re Number. The result of C_l/C_d ratio for different configurations is shown in figure 5.9. As it can be seen in figure 5.9, using vortex generators deteriorate the performance of the airfoil in low angle of attacks. After a critical angle of attack the performance of the modified airfoil is higher than that of a bare airfoil, however. The higher the Re number, the higher the critical angle.

Ultimately, an important issue needs to be addressed here and that is how a modified airfoil using double-sided VGs increases the performance of VATs. As figure 5.9 shows, lift-to-drag ratio is deteriorated before stall region although this ratio is improved in the stall and post-stall regions. This can be seen from another point of view; the

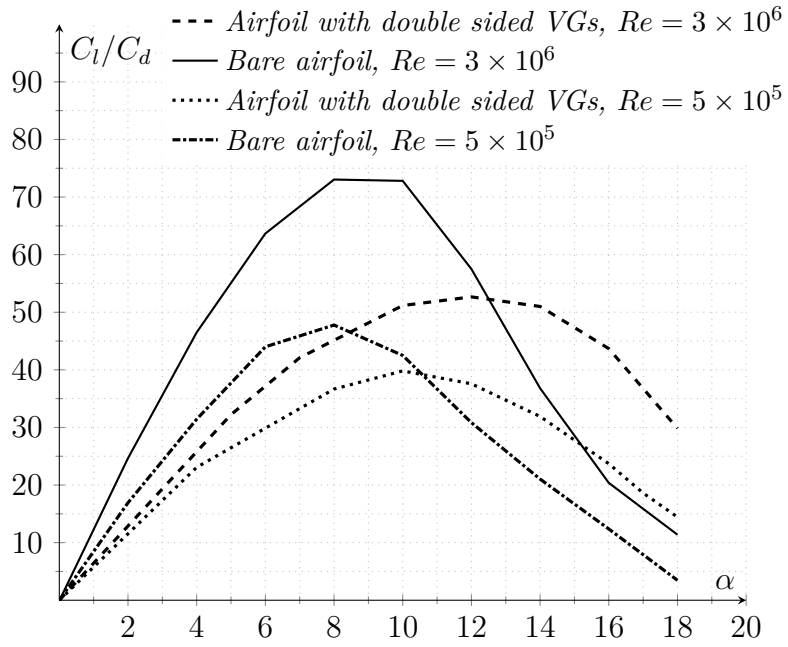


Figure 5.9: Comparison of lift to drag ratio between the bare airfoil and the passive airfoil with double-sided VGs on both suction and pressure sides

presence of VGs shifts the maximum of lift-to-drag ratio to higher angle of attacks. Therefore, if the blades work predominately in higher angles, as the blade of the technology of interest in this study and discussed in section 1.2 do, presence of VGs and this shift benefit the blades. Hence, usage of double sided VGs on the pressure and the suction sides of the blades is justified for turbines with high solidity and low tip speed ratios.

5.3 Undulating leading edge

Leading edge undulation of an airfoil is characterized using two non-dimensional parameters in the literature of the subject; non-dimensional wavelength and non-dimensional

amplitude which are defined as follows.

$$\overline{\lambda} = \frac{\lambda}{c} \tag{5.1}$$

$$\overline{A} = \frac{A}{c} \tag{5.2}$$

The curvature of the leading edge is then modified based on the following equation:

$$x_l = \overline{A} cos(2\pi z/\overline{\lambda}) \tag{5.3}$$

The schematic of the modified airfoil with leading edge tubercles is shown in figure 5.10 from the top and the front views.

In the present study, it is found that aerodynamics performance of the airfoil highly

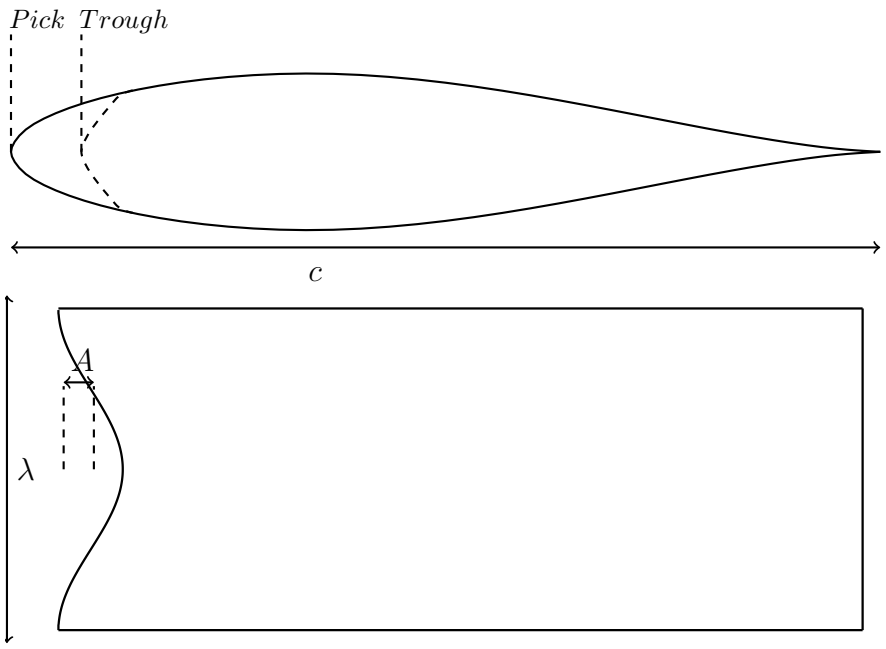


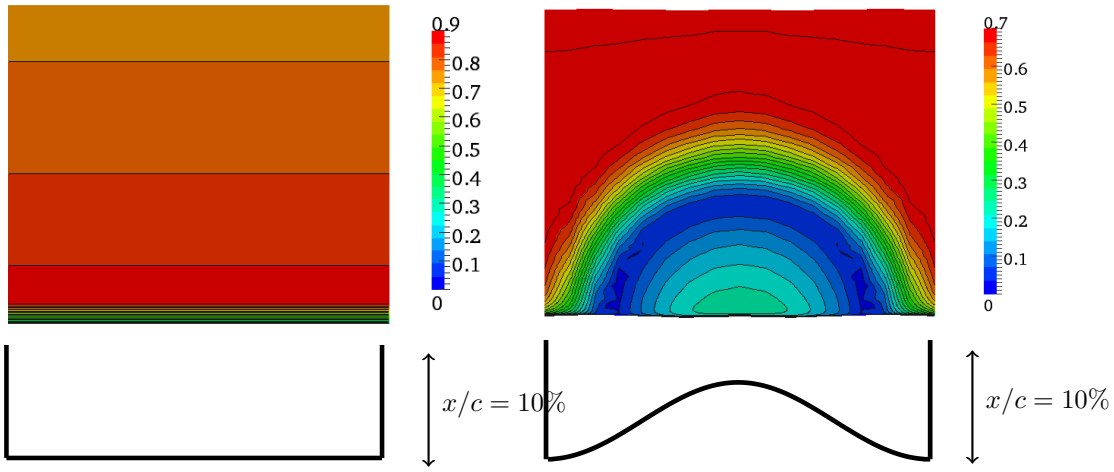
Figure 5.10: Top and front view schematics of the 3D airfoil with undulating leading edge

depends on the choice of $\overline{\lambda}$ and $\overline{A}$. Based on these control parameters and the profile

of the airfoil itself, different researchers reported different results. Results of the study published by Favier et al. [7] indicate that at $Re = 800$ and for $\overline{\lambda} = 1$ and $0.03 < \overline{A} < 0.07$, applying the leading edge tubercles decrease the drag coefficient, but decrease the lift coefficient as well. On the other hand, Hansen et al. [15] reported that applying sinusoidal leading edge protrusions, when $0.11 < \overline{\lambda} < 0.86$ and $0.03 < \overline{A} < 0.11$, increases the lift coefficient in the post stall region.

Velocity profiles obtained from the present study at a section 10% of the chord from the leading edge for $\overline{\lambda} = 0.1$ and $\overline{A} = 0.0125$ are demonstrated in figure 5.11.

The behavior of velocity profiles shown in this figure for the modified airfoil results



(a) Bare airfoil (b) Airfoil with leading edge tubercles

Figure 5.11: Comparison of the velocity profiles (in m/s) at $\alpha = 18°$ between the bare airfoil and the modified airfoil with leading edge tubercles with $\overline{\lambda} = 0.1$ and $\overline{A} = 0.0125$ at a section 10% of the chord from the leading edge

from the vorticies that are created in the x direction along the chord of the airfoil corresponding to leading edge picks as is shown in figure 5.10. The vorticity profile at this section is demonstrated in figure 5.12. On the other hand, as far as the bare airfoil is concerned, no vortices are created except in the spanwise direction due to creation of vorticies in boundary layer in the vicinity of the airfoil, yet presence of

LETs leads to non-zero vorticity in the streamline direction.

The velocity profiles of the bare airfoil and the modified airfoil at $\alpha = 18°$ are shown

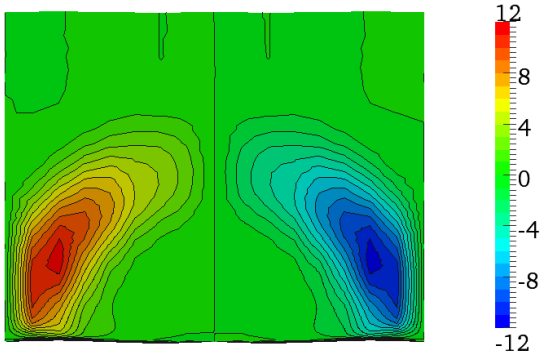


Figure 5.12: Vorticity profile at $\alpha = 18°$ for the modified airfoil with leading edge tubercles with $\overline{\lambda} = 0.1$ and $\overline{A} = 0.0125$, at a section 10% of the chord from the leading edge

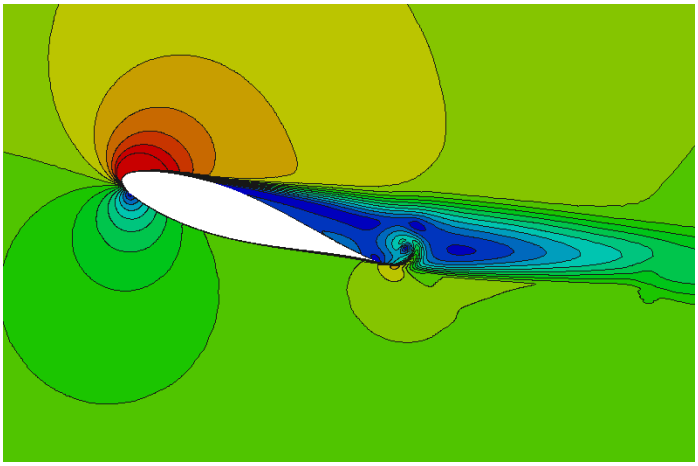
in 5.13.

It can be seen that boundary layer separates per-maturaly in the modified airfoil and this causes the lift coefficient to be lower than that of the bare airfoil.

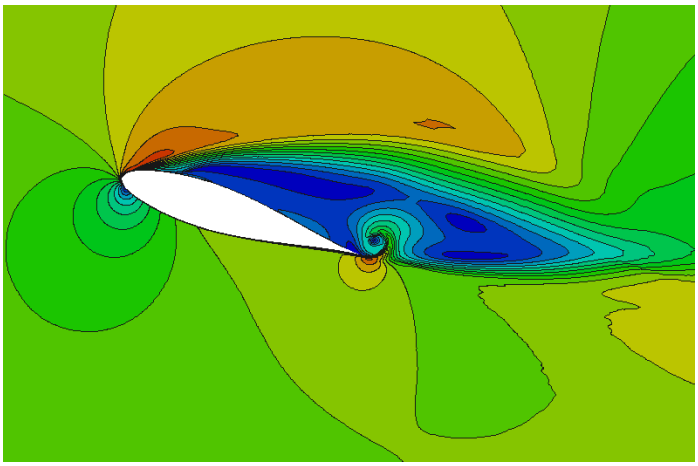
In the present study, in order to find the optimized configuration of the control parameters, $\overline{\lambda}$ and $\overline{A}$, for this particular airfoil, different configurations are tested. These configurations are presented in table 5.3

Based on table 5.3, it can be concluded that increasing the amplitude of tubercles when $1 < \overline{\lambda} < 2$ is applied, decreases the performance. Moreover, it can be seen that lower $\overline{\lambda}$ for this particular airfoil leads to better aerodynamic performance. Moreover, it can be concluded that the higher the λ/A the higher the performance. This is correct up to a certain point where increasing the amplitude does not lead to a better performance, as is shown in table 5.3 in case 12 and 13. This conclusion is in agreement with the conclusion made by Hansen et al. [15].

The optimal configuration of the control parameter , λ and A, found in table 5.3 (the last case) is then applied to the airfoil at the range of angle of attacks from $\alpha = 0°$ to



(a) Bare airfoil



(b) Airfoil with leading edge tubercles

Figure 5.13: Comparison of the velocity profiles at $\alpha = 18°$ between the bare airfoil and the modified airfoil with leading edge tubercles

$\alpha = 20°$ in order to investigate the effect of applying leading edge tubercles on the aerodynamics performance of the airfoil over the pre-stall and stall regions.

Figure 5.14 shows the effect of using tubercles with $\overline{\lambda} = 0.11$ and $\overline{A} = 0.0125$ on the performance of the airfoil at $Re = 5 \times 10^5$ for different angle of attacks. As it can be seen from figure 5.14, the lift coefficient of the modified airfoil is less than that of the bare airfoil and the drag coefficient of the modified foil is greater than that of the bare foil. This result is in agreement with the results of the study by Stein and Murray [35]

Case	$\overline{\lambda}$	$\overline{A}$	λ/A	C_l	C_d
Bare airfoil	–	–	–	1.23	0.078
2	1	0.02	0.02	0.85	0.14
3	1	0.05	0.05	0.8	0.15
4	1	0.07	0.07	0.65	0.15
6	1	0.09	0.09	0.6	0.17
7	2	0.07	0.035	0.75	0.14
8	2	0.09	0.045	0.73	0.15
9	0.4	0.01	0.025	0.8	0.22
10	0.2	0.01	0.05	0.85	0.22
11	0.2	0.015	0.075	0.75	0.14
12	0.1	0.015	0.15	0.75	0.14
13	0.11	0.0125	0.113	0.9	0.16

Table 5.3: Effect of the tubercles' control parameters on the lift and the drag coefficients at AOA=16° and $Re = 5 \times 10^5$

and disagreement with the results of the study by Hansen et al. [15]. As it was stated before, the study by Murray showed that [26] changing the foil profile to the one used by Miklosovic et al. [25] in a swept wing, results in greater performance in comparison to the same model with an unswept leading edge. This could be attributed to two different, factors. First, one can argue that using swept leading edge, which more accurately resembles the humpback whale's wing, may have changed the their results. Second, it is certainly possible that this increase in performance of the airfoil is due to the profile of the airfoil itself. Based on the results of figure 5.14 and reference [15], it can be seen that the airfoil used by Hansen et al. does go into deep stall region, yet the airfoil profile used in the present study does not go into deep stall region and does not experience drastic loss of lift over the range of angle of attacks over which the numerical results of this study are obtained.

From another point of view, results of Hansen et al. [15] indicate that the lift coefficient decreases in the stall region and flattens out in the post-stall region. This is reported

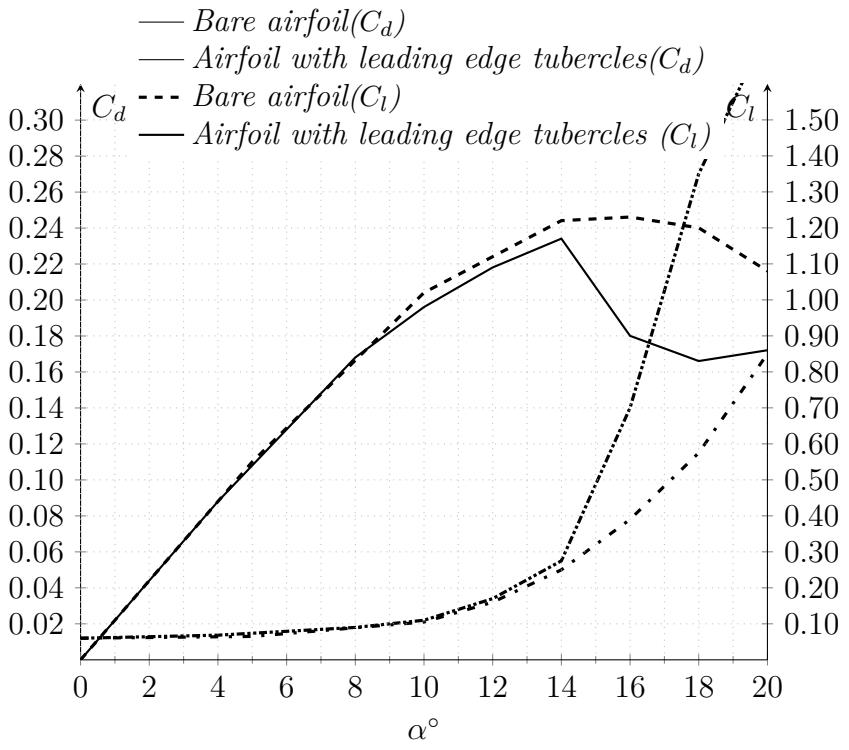


Figure 5.14: Effect of using leading edge tubercles with $\overline{\lambda} = 0.1$ and $\overline{A} = 0.0125$ at $Re = 5 \times 10^5$.

by other investigators such as the study by Nierop et al. [38].

To summarize, it results from the current study that applying leading edge tubercles does not improve the performance of all airfoil shapes. Depending on how soon the airfoil experience the deep stall and loss of lift when the angle of attack is increased, applying LETs may or may not be beneficial. The sooner the airfoil experience the drastic loss of lift, the higher the performance of the modified airfoil with LETs in comparison to unmodified one. Therefore, since the airfoil profile that is of interest in this study, NACA 63-018, does not experience drastic loss of lift in the range of angle of attacks studied here, applying LETs is not beneficial for this airfoil at this range of angle of attacks and this Re number.

Chapter 6

Conclusion and scope of future work

In this research flow control over NACA 63-018 is performed. To this end, vortex generators and leading edge tubercles are used. 3D simulations of modified airfoils are carried out using OpenFOAM, an open source CFD package, in order to explore how VGs and LETs can be implemented on a NACA-63018 airfoil to improve it's performance.

It is shown that applying vortex venerators on both the pressure side and the suction of the airfoil increases lift-to-drag ratio in higher angle of attacks. Moreover, it is shown that applying leading edge tubercles does not increase the performance of this airfoil as it does not go into deep stall region in the range of angle of attacks studied in the present study.

In heavily loaded airfoils encountered in hydro-kinetic vertical axis turbine with high solidity and low tip speed ratios, blades operate mostly in stall region, which as a consequence dramatically reduces the performance of the device. Therefore, if double sided vortex generators, as suggested in this study, are applied on the blades of the aforementioned devices, the average lift-to-drag ratio will increase and this ultimately increases the efficiency of the device.

Appendix A

SIMPLE Algorithm in OpenFOAM

A.1 Basic information

In this section the implementation of SIMPLE Algorithm in OpenFOAM is explained. To fully understand the method, it is vital to highlight some points before proceeding.
1- OpenFOAM uses collocated grids for the calculations not staggered ones. Therefore, it is required to adjust the face fluxes in order to eliminate the zigzag pattern in pressure field when collocated grids are used.
2- Rhie and Chow correction [32] is not explicitly used in OpenFOAM. However, the algorithm used in OpenFOAM ensures the oscillation free pressure field.
3- PISO and PIMPLE algorithm which are used for unsteady problems have quite similar structures in their source codes as does SIMPLE. Therefore, the notes in this appendix can be extrapolated, with minor differences, to PISO and PIMPLE algorithm.

A.2 Mathematics of SIMPLE algorithm

In this appendix it is tried to explain the SIMPLE algorithm and corresponding lines in the source code of simpleFoam algorithm. The Continuity and Navier-Stokes equations:

$$\nabla.(\rho\mathbf{U}) = \mathbf{0} \tag{A.1}$$

$$\nabla.(\mathbf{UU}) - \nabla.(\nu\nabla\mathbf{U}) = -\nabla\mathbf{p} \tag{A.2}$$

Two issues need to be addressed here; non-linearity of convection term and coupling of pressure and velocity.

Regarding the first issue, OpenFOAM linearize the convection term. To this end, the velocity of previous step is used for one of the **U** term in the convection term of the momentum equation and is regarded as a constant field. This causes a lag between the velocity and pressure. However, for steady state problem when the final solution is reached, the velocity that is being used for the face fluxes are the same as the velocity that is obtained in the next iteration. For unsteady cases, it is argued that when **Co** number is small enough, this lag does not considerably affect the accuracy of the solution[18].

The coupling of the velocity and the pressure is addressed using an iterative process and a predictor and corrector method; the velocity is predicted, and is used to update a pressure filed that satisfies the continuity equation. The velocity field is then updated using the information of the pressure field.

To begin with, the semi-discretized momentum equation for each cell is written as follow:

$$a_p \mathbf{U_p} = \sum_{\mathbf{n}} \mathbf{a_n U_n} - \nabla \mathbf{p} = \mathbf{H(U)} - \nabla \mathbf{p} => \mathbf{U_p} = \frac{\mathbf{H(U)}}{\mathbf{a_p}} - \frac{\nabla \mathbf{p}}{\mathbf{a_p}} \tag{A.3}$$

In this equation, $\mathbf{U_p}$ refers to the cell center velocity of each **p**oint. a_p, a_n and $H(\mathbf{U})$ are functions of velocity, convective flux (F) and conductance (D) of neighbor's faces. here F and D are defined similar to:

$$F = \rho \mathbf{U_f} \tag{A.4}$$

$$F = \frac{\Gamma_f}{l} \tag{A.5}$$

where Γ and l are respectively related to the diffusive term of the momentum equations (Viscose term) and the cell to cell distance of each cell and its neighbors. For more edtails about the FV discretization of NS equation reference [28] can be used. It is important to note here that $H(\mathbf{U})$ contains the effect of the convective and diffusive terms of Navier-Stokes equations while the effect of pressure gradient term is not included in $H(\mathbf{U})$. Hence, $\mathbf{U_p}$ can be written as the sum of a non-pressure part and a pressure part.

$$\mathbf{U_p} = \mathbf{U_{np}} - \mathbf{U_{pp}} \tag{A.6}$$

Where:

$$\mathbf{U_{np}} = \frac{\mathbf{H(U)}}{\mathbf{a_p}} \tag{A.7}$$

$$\mathbf{U_{pp}} = \frac{\nabla \mathbf{p}}{\mathbf{a_p}} \tag{A.8}$$

OpenFOAM uses a special classes to obtain $H(\mathbf{U})$ and a_p from the discretized Navier-Stokes equations. Assuming that the discretized Navier-Stokes equations can be written in a Matrix format below:

$$C\mathbf{U} = R \tag{A.9}$$

Where $\mathbf{U}$ is a matrix of cell center velocities, C and R are respectively the coefficient matrix and residual matrix when the effect of the pressure gradient term is not included. In order to obtain the a_p matrix the following operations are conducted on equation A.9.

$$(A + T)\mathbf{U} = R \tag{A.10}$$

$$AU = R - TU => AU = H \tag{A.11}$$

$$AU = H \tag{A.12}$$

Here A is a diagonal matrix which constructs the a_p coefficients, and H will resemble the remaining terms in $H(\mathbf{U})$ in equation A.7.

Moreover, there is also another point that should be made before proceeding to the algorithm and that is about the implicit and explicit discretization of terms in OpenFOAM. OpenFOAM uses fvc, which stands for Finite Volume Calculus, for explicit discretization and fvm,which stands for Finite Volume Method, for implicit discretization.

As it was stated, continuity equation is used to obtain a pressure filed that satisfy the continuity. In order to obtain the pressure equation, the following equations are used. Using the divergence theorem, the continuity equation is correlated to face velocities as follows:

$$\nabla.U = \sum_f \mathbf{U_f}.\mathbf{S_f} = \mathbf{0} \tag{A.13}$$

In order to obtain face center velocities, equation A.3 is interpolated to face center and gives:

$$\mathbf{U_f} = \left(\frac{\mathbf{H(U)}}{\mathbf{a_p}}\right)_\mathbf{f} - \left(\frac{\mathbf{1}}{\mathbf{a_p}}\right)_\mathbf{f}\left(\nabla\mathbf{p}\right)_\mathbf{f} \tag{A.14}$$

Using the second term of equation A.13 it follows:

$$\sum_f \mathbf{U_f}.\mathbf{S_f} = \sum_f \left(\left(\frac{\mathbf{H(U)}}{\mathbf{a_p}}\right)_\mathbf{f} - \left(\frac{\mathbf{1}}{\mathbf{a_p}}\right)_\mathbf{f}\left(\nabla\mathbf{p}\right)_\mathbf{f}\right)\mathbf{S_f} = \mathbf{0} \tag{A.15}$$

$$\sum_f \left(\frac{H(\mathbf{U})}{a_p}\right)_f .\mathbf{S_f} = \sum_f \left(\frac{1}{a_p}\right)_f \left(\nabla p\right)_f .\mathbf{S_f} \tag{A.16}$$

And using the divergence theorem,

$$\nabla.\left(\frac{H(\mathbf{U})}{a_p}\right) = \nabla.\left(\left(\frac{1}{a_p}\right)\left(\nabla p\right)\right) \tag{A.17}$$

$$\nabla.\left(\frac{H(\mathbf{U})}{a_p}\right) = \nabla^2\left(\frac{p}{a_p}\right) \tag{A.18}$$

Which is the pressure equation that OpenFOAM uses to update the pressure field.

SIMPLE algorithm

With all the information explained above, the implementaion of the SIMPLE algorithm can be expressed. The SIMPLE algorithm is written as follows [28, 9, 18]:

- 1) Guess the pressure field.
- 2) Update the intermediate velocity field $\mathbf{U}^*$ based on the pressure filed using the momentum equation. In this step the pressure filed is known while the velocity field in unknown. Therefore, discretization of the pressure filed is explicit (uses the information that is available) and discretization of the remaining terms (velocity filed) is implicit (uses the information of the next iteration which is not available). Since implicit method is used here, the procedure to find the velocity will be iterative.
- 3) Calculate the non-pressure part of the velocity, $H(\mathbf{U})/\mathbf{a_p}$, and the face fluxes using this part. This part is used in the pressure equation.
- 4) Solve the pressure equation to obtain the pressure field. In this part, the pressure discretization in implicit and discretization of the U_{np} is explicit.

- 5) Correct the velocity fluxes so that the effect of the pressure part of the velocity is taken into account.

- 6) Under-relax the pressure field and correct the velocity by subtracting the pressure part of the velocity, U_{pp}, from the non-pressure part.

- 7) Go back to step 3 and perform the same loop until the solution is reached within the desirable tolerance.

A.3 OpenFOAM source code

In this section, implementation of the seven steps described in section A.2 are explained in the source code of OpenFOAM.

- 1) At the beginning of the source code the left hand side of equation A.2 is constructed implicitly. This is achieved using fvm syntax. Here, the first U in the convective term of equation A.2 is substituted by phi which is the face flux of the previous velocity. This is how linearizion of the convective term is performed in OpenFOAM.

  tmp<fvVectorMatrix>UEqn
  (
  fvm::div(phi, U) - fvm::laplacian(nu, U)
  );

  which is equal to :

$$\nabla.(\mathbf{UU}) - \nabla.(\nu\nabla\mathbf{U})$$

- 2) Under-relax the equation for U,

UEqn.relax();

And solve the momentum predictor. As explained in step 2 of section A.2, the left hand side of equation A.2 is implicitly discretized and the right hand side explicitly.

solve (UEqn == -fvc::grad(p));

which is equal to :

$$\nabla.(\mathbf{UU}) - \nabla.(\nu\nabla\mathbf{U}) = -\nabla\mathbf{p}$$

Update the boundary conditions for p:

p.boundaryField().updateCoeffs();

- 3) Construct the **AU**, which contains the a_p coefficients, $H(\mathbf{U})$ and calculate the non-pressure part of the velocity $\frac{H(\mathbf{U})}{a_p}$

volScalarField AU = UEqn().A();
U = UEqn().H()/AU;
UEqn.clear();

Here, **UEqn().H()** constructs the $H(\mathbf{U})$ and **UEqn().A()** the a_p. All these result in $\mathbf{U_{np}} = \frac{\mathbf{H(U)}}{\mathbf{a_p}}$.
Finally, calculate the flux using the non-pressure part of the velocities:

phi = fvc::interpolate(U) & mesh.Sf();
adjustPhi(phi, U, p);

which calculates the flux at the cell face and adjusts the flux at the boundaries.

- 4) Define and solve the pressure equation and repeat for the prescribed number of non-orthogonal corrector steps.

  The left hand side of the pressure equation (equation A.18) is the divergence of the non-pressure part of the velocity. The right hand side of the pressure equation A.18 simply is the laplacian of pressure. The left hand side is discretized explicitly and the right hand side implicitly. Hence:

  fvScalarMatrix pEqn
  (
  fvm::laplacian(1.0/AU, p) == fvc::div(phi)
  );
  pEqn.setReference(pRefCell, pRefValue);
  pEqn.solve();

  which is translated as:

$$\nabla^2\left(\frac{p}{a_p}\right) = \nabla.\left(\frac{H(\mathbf{U})}{a_p}\right)$$

- 5) The pressure filed is updated. However, the flux which has been used should be corrected so that the effect of the pressure part of the velocity is included in the final flux. Correct the flux using the calculated flux of the pressure part of the velocity:

  phi -= pEqn.flux();

  Here, **pEqn.flux()** calculates the flux of the pressure part of the velocity, meaning $\sum_f \left(\frac{\nabla \mathbf{p}}{\mathbf{a_p}}\right)_\mathbf{f}.\mathbf{S_f}$. This flux needs to be subtracted from the non-pressure

part of the flux, $\sum_f \left(\frac{\mathbf{H(U)}}{\mathbf{a_p}}\right)_\mathbf{f}.\mathbf{S_f}$ so that the corrected flux is obtained and satisfies the continuity equation A.15. In other words:

$$\phi = \phi_{np} - \phi_p$$

$$\sum_f \mathbf{U_f}.\mathbf{S_f} = \sum_\mathbf{f} \left(\frac{\mathbf{H(U)}}{\mathbf{a_p}}\right)_\mathbf{f}.\mathbf{S_f} - \sum_\mathbf{f} \left(\frac{\nabla \mathbf{p}}{\mathbf{a_p}}\right)_\mathbf{f}.\mathbf{S_f}$$

- 6) Under-relax the pressure for the momentum corrector and apply the velocity correction since velocity should be corrected as well:

  p.relax();

  U -= fvc::grad(p)/AU;

  U.correctBoundaryConditions();

  The second line is translated as:

  $$U_p = U_{np} - U_{pr}$$
  $$\overrightarrow{U_p} = \frac{H(\overrightarrow{U})}{a_p} - \frac{\nabla p}{a_p}$$

  which gives the corrected velocity.

- 7) repeat the above procedure until the residuals meet the desirable tolerance. The residuals are calculated using:

  eqnResidual = solve (UEqn() == -fvc::grad(p)).initialResidual();

Appendix B

An OpenFOAM Tutorial

B.1 Structure of the case

In this appendix a basic OpenFOAM structure of the simulations discussed in the present study is presented. It is tried to present the case as simple as possible so that it can be used by OpenFOAM community.

The tutorial has three directory, 0 folder, constant folder and system folder as is shown in the following figure. The Boundary conditions of the model are presented in U, p, k, omega and nut. The computational domain is created in the constant/polyMesh/blockMesh.m4. The files under the constant folder select the turbulent model and physical properties of the fluid. Finally, under system folder, the dictionaries required for mesh manipulation, parallelization and control of the solution are presented. The files on the root folder of the case can be used to monitor the lift coefficient, the drag coefficient and the residuals. Ultimately, the case can be run by executing the following command in the terminal.

./clean in order to clean the case.

./mesh to create the computational domain.

./Allrun to run the case.

B.2 Files

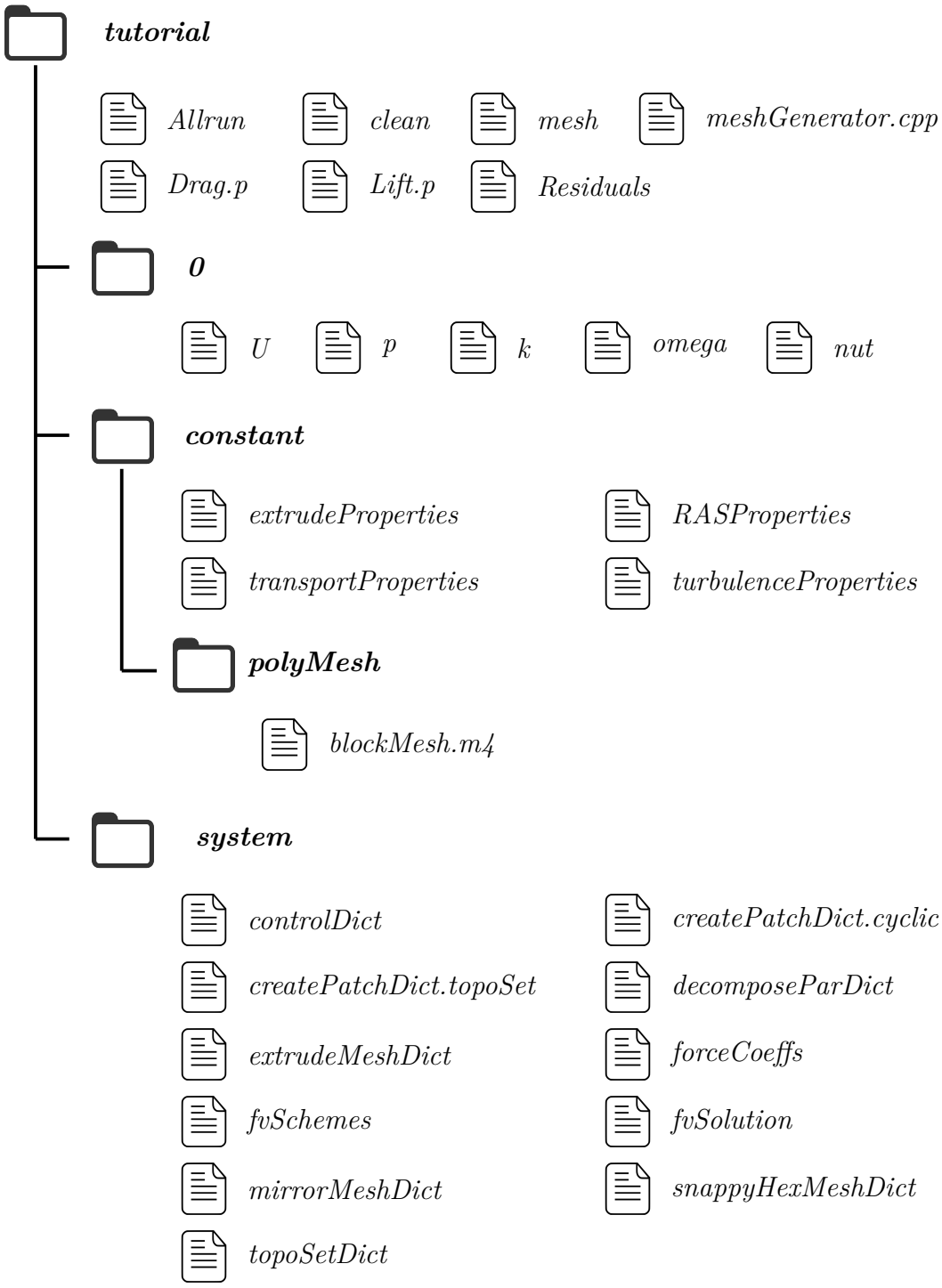


Figure B.1: Structure of the OpenFOAM case

Bibliography

[1] Drag components. http://goo.gl/2Fmm9G. Accessed: 2014-05-16.

[2] pimplefoam. http://goo.gl/6ANQU0. Accessed: 2014-05-16.

[3] Ira H Abbott and Albert E Von Doenhoff. *Theory of wing sections: including a summary of airfoil data*. Courier Dover Publications, 2012.

[4] John David Anderson. *Introduction to flight*, volume 199. McGraw-Hill Boston, 2005.

[5] Prasun Chatterjee and Raymond N Laoulache. Performance modeling of ducted vertical axis turbine using computational fluid dynamics. *Marine Technology Society Journal*, 47(4), 2013.

[6] Laurence Joseph Clancy. *Aerodynamics*. Pitman London, 1975.

[7] Julien Favier, Alfredo Pinelli, and Ugo Piomelli. Control of the separated flow around an airfoil using a wavy leading edge inspired by humpback whale flippers. *Comptes Rendus Mecanique*, 340(1):107–114, 2012.

[8] Carlos Simao Ferreira, Gerard Van Bussel, Fulvio Scarano, and Gijs van Kuik. 2d piv visualization of dynamic stall on a vertical axis wind turbine. In *45th AIAA Aerospace Sciences Meeting and Exhibit/ASME Wind Energy Symposium*, 2007.

[9] Joel H Ferziger and Milovan Perić. *Computational methods for fluid dynamics*, volume 3. Springer Berlin, 2002.

[10] Frank E Fish, Paul W Weber, Mark M Murray, and Laurens E Howle. The tubercles on humpback whales' flippers: Application of bio-inspired technology. *Integrative and comparative biology*, 51(1):203–213, 2011.

[11] Mohamad Gad-el Hak. Flow control. *Applied mechanics reviews*, 42(10):261–293, 1989.

[12] Mohamed Gad-el Hak. Modern developments in flow control. *Applied Mechanics Reviews*, 49(7):365–379, 1996.

[13] Benjamin Gal-Or. *Vectored Propulsion, Supermaneuverability and Robot Aircraft*. Springer, 1990.

[14] Gilles Godard and Michel Stanislas. Control of a decelerating boundary layer. part 1: Optimization of passive vortex generators. *Aerospace Science and Technology*, 10(3):181–191, 2006.

[15] AC Hansen and CP Butterfield. Aerodynamics of horizontal-axis wind turbines. *Annual Review of Fluid Mechanics*, 25(1):115–149, 1993.

[16] Kristy L Hansen, Richard M Kelso, and Bassam B Dally. Performance variations of leading-edge tubercles for distinct airfoil profiles. *AIAA journal*, 49(1):185–194, 2011.

[17] Eastman N Jacobs, Kenneth E Ward, and Robert M Pinkerton. The characteristics of 78 related airfoil sections from tests in the variable-density wind tunnel. Technical report, DTIC Document, 1933.

[18] Hrvoje Jasak. Error analysis and estimation for the finite volume method with applications to fluid flows. 1996.

[19] JC Lin, GV Selby, and FG Howard. Exploratory study of vortex-generating devices for turbulent flow separation control. *AIAA paper*, (1991-0042), 1991.

[20] John C Lin. Review of research on low-profile vortex generators to control boundary-layer separation. *Progress in Aerospace Sciences*, 38(4):389–420, 2002.

[21] YF Lin, K Lam, L Zou, and Y Liu. Numerical study of flows past airfoils with wavy surfaces. *Journal of Fluids and Structures*, 36:136–148, 2013.

[22] Pier Marzocca. The naca airfoil series. *Clarkson University, March*, 2009.

[23] Barnes Warnock McCormick. *Aerodynamics of V/STOL flight*. Courier Dover Publications, 1967.

[24] Florian R Menter. Two-equation eddy-viscosity turbulence models for engineering applications. *AIAA journal*, 32(8):1598–1605, 1994.

[25] DS Miklosovic, MM Murray, LE Howle, and FE Fish. Leading-edge tubercles delay stall on humpback whale (megaptera novaeangliae) flippers. *Physics of Fluids (1994-present)*, 16(5):L39–L42, 2004.

[26] MM Murray, DS Miklosovic, FE Fish, and LE Howle. Effects of leading edge tubercles on a representative whale flipper model at various sweep angles. In *Durham, NH: Conference Proceedings of the 14th International Symposium on Unmanned Untethered Submersible Technology*, 2005.

[27] Ion Paraschivoiu. *Wind turbine design: with emphasis on Darrieus concept*. Presses inter Polytechnique, 2002.

[28] Suhas Patankar. *Numerical heat transfer and fluid flow*. CRC Press, 1980.

[29] HH Pearcey. *Shock-induced separation and its prevention by design and boundary layer control*. Pergamon, 1961.

[30] Charles S Peskin. The immersed boundary method. *Acta numerica*, 11:479–517, 2002.

[31] DM Rao and TT Kariya. Boundary-layer submerged vortex generators for separation control-an exploratory study. *AIAA paper*, 3546:1988, 1988.

[32] CM Rhie and WL Chow. Numerical study of the turbulent flow past an airfoil with trailing edge separation. *AIAA journal*, 21(11):1525–1532, 1983.

[33] Hua Shan, Li Jiang, Chaoqun Liu, Michael Love, and Brant Maines. Numerical study of passive and active flow separation control over a naca0012 airfoil. *Computers & Fluids*, 37(8):975–992, 2008.

[34] Mitsuhiro Shiono, Katsuyuki Suzuki, and Seiji Kiho. An experimental study of the characteristics of a darrieus turbine for tidal power generation. *Electrical Engineering in Japan*, 132(3):38–47, 2000.

[35] B Stein and MM Murray. Stall mechanism analysis of humpback whale flipper models. *Proceedings of Unmanned Untethered Submersible Technology (UUST), UUST05*, 5, 2005.

[36] Florian Von Stillfried, Stefan Wallin, and Arne V Johansson. Vortex-generator models for zero-and adverse-pressure-gradient flows. *AIAA journal*, 50(4):855–866, 2012.

[37] HD Taylor. The elimination of diffuser separation by vortex generators. *United Aircraft Corporation Connecticut, Research Department Report No R-4012-3*, 1947.

[38] Ernst A van Nierop, Silas Alben, and Michael P Brenner. How bumps on whale flippers delay stall: an aerodynamic model. *Physical review letters*, 100(5):054502, 2008.

[39] Gurudutt Venkatesh. *Efficiency Analysis of Vertical Axis Hydokinetic Turbine Using Computational Fluid Dynamics: A Thesis in Mechanical Engineering*. PhD thesis, University of Massachusetts, Dartmouth, 2011.

[40] Florian von Stillfried. *Computational fluid-dynamics investigations of vortex generators for flow-separation control*. PhD thesis, KTH, Turbulence, 2012. QC 20120511.

[41] Florian von Stillfried, Stefan Wallin, and AV Johansson. Statistical modeling of the influence of turbulent flow separation control devices. In *47th AIAA Aerospace Sciences Meeting, Orlando, Florida*, pages 2009–1501, 2009.

[42] P Watts and FE Fish. The influence of passive, leading edge tubercles on wing performance. In *Proc. Twelfth Intl. Symp. Unmanned Untethered Submers. Technol.* Auton. Undersea Syst. Inst. Durham New Hampshire, 2001.